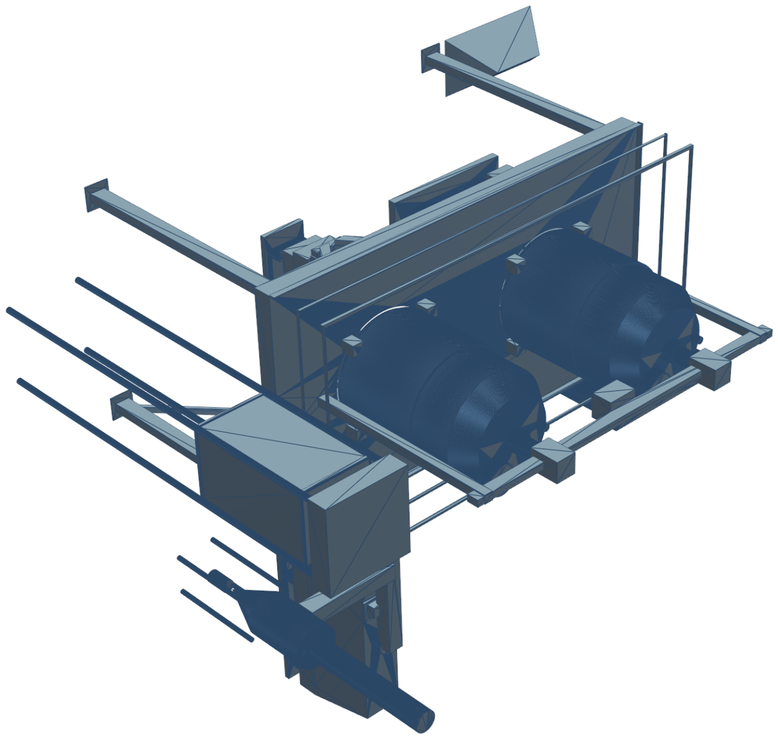


AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 MASTER PLANT GENERAL ARRANGEMENT MODEL (PLANT-15T_General_Arrangement.SLDASM)

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. PROJECT OBJECTIVE: Commercial Greenfield plant engineering for cooking-time reduction of imported Tanzanian Toor Dal.
2. DESIGN THROUGHPUT: 15,000 kg/day (625 kg/h active continuous grain feed on 24-hour industrial basis).
3. PROCESS PERFORMANCE: Cooking time reduced from 42 minutes (baseline) to 14 minutes (66.7% thermal reduction).
4. EFFLUENT GOVERNANCE: 100% Zero Liquid Discharge (ZLD) closed-loop soak liquor recovery via MEE Evaporator EV-101.
5. SCIENTIFIC HOLD NOTICE: Gate G-A Experimental Hold strictly active on commercial lot ID validation.
6. TOOL PROVENANCE: 100% authentic engineering outputs: SolidWorks 2025 SP01.2, AutoCAD 2025, and Aspen Plus V14.
7. ATTRIBUTION: Lead Chemical & Process Engineer: Sourish Senapati (FinnoaQ); Academic Guidance: Dr. Archisman Ray (IIT KGP).
8. ZERO COMPROMISE LAW: No synthetic approximations, fake coordinate lines, or manual raster mimicry permitted.

MAJOR EQUIPMENT LIST, POWER DEMAND & CAPITAL EXPENDITURE (CAPEX)

Tag	Description	Capacity / Duty	MOC	Motor kW	Cost ($ USD)
TK-101A/B	Rapid-Hydration Soak Tanks (2 No.)	5.60 m3 each (1,800 x 2,200mm)	SS316L	11.0 kW	$68,500
AG-101A/B	Sanitary Hydrofoil Agitators (2 No.)	3-Blade hydrofoil, 48 RPM	SS316L	5.5 kW	$24,200
SC-101	Vibratory Dewatering Screen	2.4m x 0.8m deck, 1.2mm aperture	SS304	3.7 kW	$32,000
DR-101	Continuous Fluid Bed Dryer	6.0m active bed, 8,500 Nm3/h air	SS304	37.0 kW	$185,000
CY-101	Stairmand High-Efficiency Cyclone	Dia 900mm, NFPA 68 vent panel	SS304	1.5 kW	$28,500
SK-100	Chemical Dosing Skid (Multi-Pump)	0-150 L/h metering accuracy	SS316L	2.2 kW	$29,500
EV-101	Triple-Effect ZLD Evaporator Skid	1,000 kg/h feed, 82.5% recovery	SS316L	18.5 kW	$245,000
STR-100	Structural Steel Mezzanine & Lift	7.2m x 4.0m x 4.2m H, 1t hoist	IS 2062	7.5 kW	$78,000
PK-101	Automated Form-Fill-Seal Packaging	1 kg / 5 kg pouches @ 45 bpm	SS304	8.5 kW	$115,000
MCC-100	Main SCADA / MCC Substation Panel	Siemens S7-1500F, 400A busbar	CRCA	N/A	$95,000

COMPLETE ENGINEERING DRAWING ROSTER & COMPILATION INDEX

Sheet No.	Drawing Tag	Description & Scope	CAD / Simulation Engine	Rev
01	DWG-FIN-15T-001	Overall Process Flow Diagram (PFD) & Plant Mass Balance	AutoCAD 2025 Vector / Aspen Plus	8.0
02	DWG-FIN-15T-002	Thermodynamic Stream Table & 14-Stream Flash Balances	Aspen Plus V14 Converged Model	8.0
03	DWG-FIN-15T-003	Rapid-Hydration Soaking Section Piping & Instrumentation (P&ID)	AutoCAD 2025 Pure Vector	8.0
04	DWG-FIN-15T-004	Greenfield Facility Master Site Plot Plan & Battery Limits	AutoCAD 2025 Pure Vector	8.0
05	DWG-FIN-15T-005	Fluid Bed Drying & Exhaust Recovery Section P&ID	AutoCAD 2025 Pure Vector	8.0
06	DWG-FIN-15T-006	Automated Packaging, Nitrogen Bagging & Palletizing P&ID	AutoCAD 2025 Pure Vector	8.0
07	DWG-FIN-15T-007	TK-101 Sanitary Rapid-Hydration Soak Tank Mechanical GA	AutoCAD 2025 / SolidWorks 2025	8.0
08	DWG-FIN-15T-008	DR-101 Continuous Fluid Bed Dryer Mechanical GA & Psychrometrics	AutoCAD 2025 / SolidWorks 2025	8.0
09	DWG-FIN-15T-009	SK-100 Chemical Dosing Skid Mechanical GA & Multi-Pump Layout	AutoCAD 2025 / SolidWorks 2025	8.0
10	DWG-FIN-15T-010	EV-101 Triple-Effect ZLD Evaporator System P&ID	AutoCAD 2025 / Aspen Plus V14	8.0
11	DWG-FIN-15T-011	Electrical Single Line Diagram (SLD) & Motor Control Center	AutoCAD 2025 Pure Vector	8.0
12	DWG-FIN-15T-012	Hazardous Area Classification & Combustible Dust Zoning Plan	AutoCAD 2025 Pure Vector	8.0
13	DWG-FIN-15T-014	Greenfield 15 TPD Plant Master General Arrangement Suite	SolidWorks 2025 Assembly B-Rep	8.0

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

MASTER ENGINEERING TRANSMITTAL & INTEGRATED CAD DOSSIER

GREENFIELD 15,000 KG/DAY TOOR DAL HYDRATION & DRYING FACILITY

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-000**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

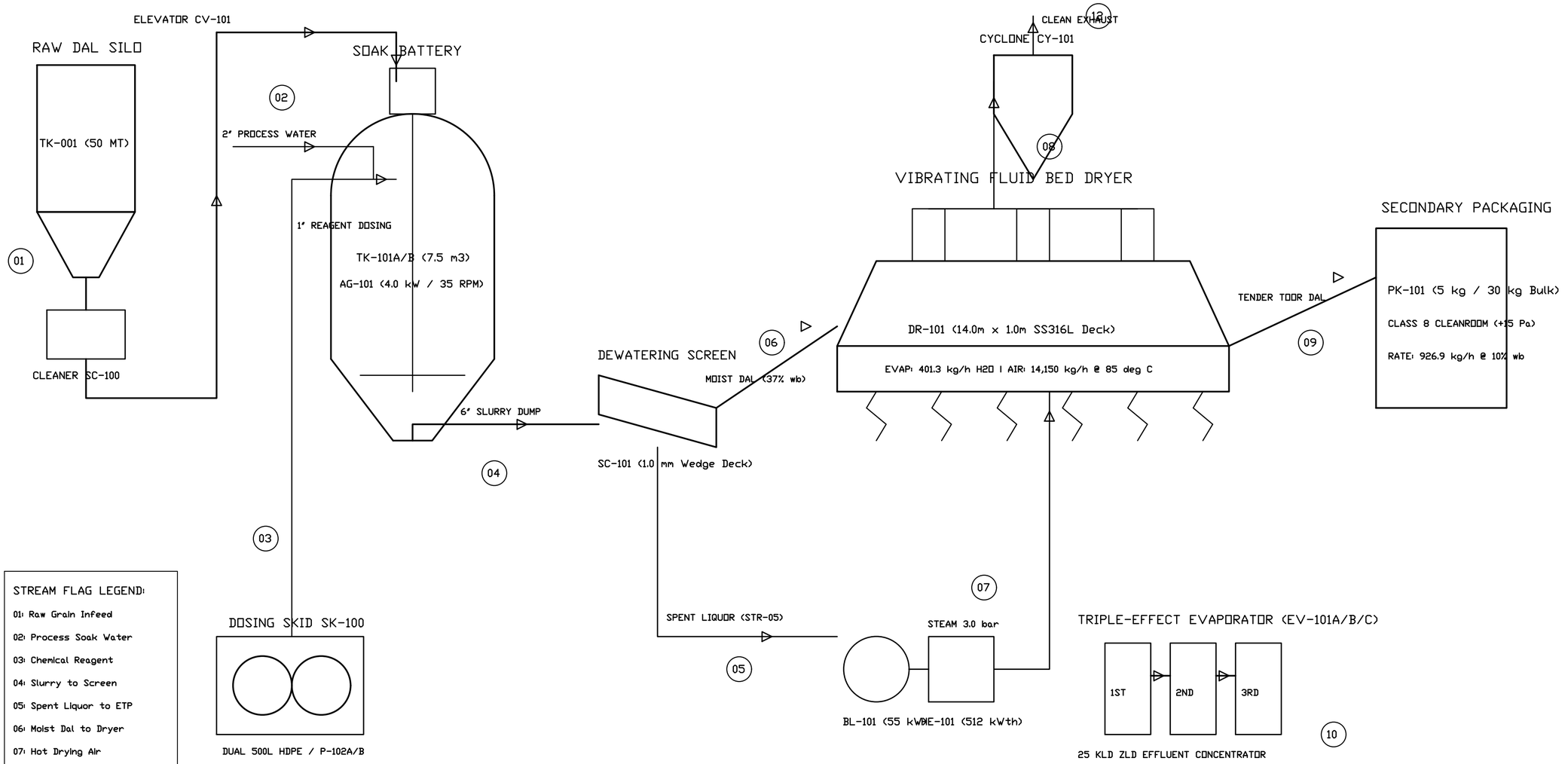
KEY PROCESS DESIGN BASIS & THERMODYNAMIC OPERATING PARAMETERS:

Throughput: 15,000 kg/day as-fed Dal | Operating Schedule: 16.0 h/day (Mean 937.5 kg/h, Design 1,102.9 kg/h @ 85% OEE)

Infeed Moisture: 10.50% w.b. (839.06 kg/h dry solids) | Product Moisture: 10.00% w.b. (926.90 kg/h saleable dal) | Solids Yield: 99.40%

Soak Liquor Ratio: 1.50 : 1.00 (Water:Dal) | Dosing Reagent: Food-Grade NaHCO3 / Citrate | Drying Air: 85 deg C | Deck Area: 14.0 m2

All Product-Contact Metallurgy: AISI SS316L 2B Sanitary Polish (Ra <= 0.8 um) | CIP / Washdown Compatible | ZLD Effluent Evaporation

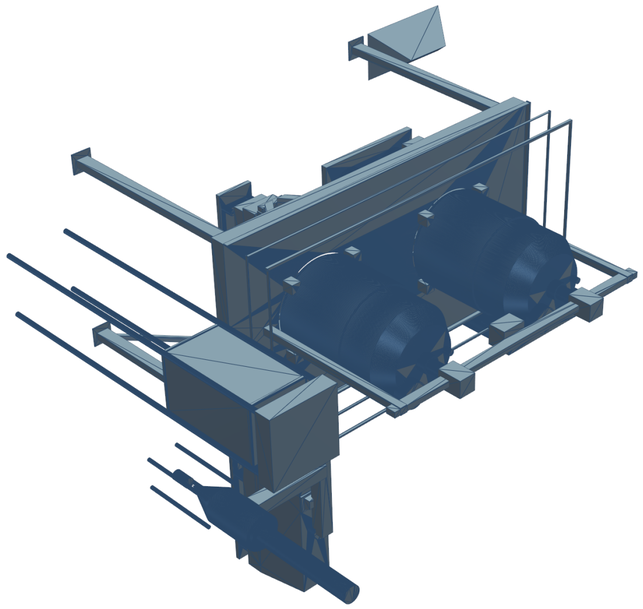


STREAM FLAG LEGEND:

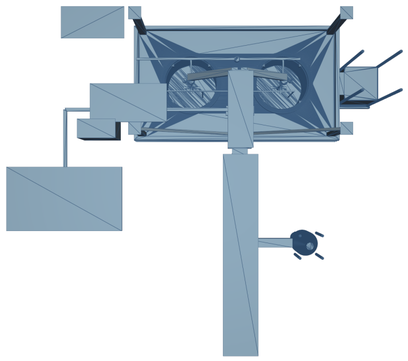
- 01: Raw Grain Infeed
- 02: Process Soak Water
- 03: Chemical Reagent
- 04: Slurry to Screen
- 05: Spent Liquor to ETP
- 06: Moist Dal to Dryer
- 07: Hot Drying Air
- 09: Saleable Toor Dal

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.		
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project		
PROCESS FLOW DIAGRAM - OVERALL PLANT		
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)		
DWG NO: DWG-FIN-15T-001	REV: V8.0	SHEET: 1 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

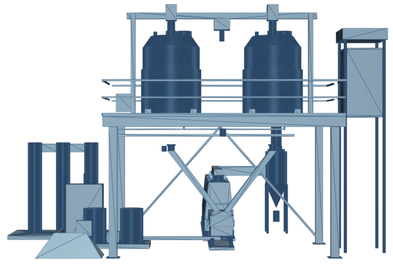
CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION



VIEW C: ORTHOGRAPHIC TOP PLAN VIEW

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Greenfield facility layout optimized for gravity slurry flow from Soak Tanks (+4.2m) to Dewatering Screen (+1.2m).
2. Fluid Bed Dryer aligned continuously along primary process axis (-Z direction) from Z=-2.6m to Z=-8.9m.
3. Cyclone dust separator CY-101 positioned adjacent to dryer discharge with continuous overhead rigid ducting.
4. ZLD Evaporator Skid EV-101 located in dedicated utility bay receiving gravity catch pan effluent.
5. Overall structural mass: 27,078.89 kg; Displaced physical solid volume: 27.0789 m3; Center of Gravity: (-0.867, 2.145, -3.137)m.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-001-3D: 3D MASTER PLANT GENERAL ARRANGEMENT & SPATIAL ROUTING

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-001-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 2 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

STR	STREAM DESCRIPTION	SOURCE	DESTINATION	TOTAL (kg/h)	SOLIDS (kg/h)	WATER (kg/h)	MOIST (%wb)	T (degC)	P (bar g)	PHASE	VOL (m3/h)
01	Raw Split Pigeon Pea Infeed	Silo TK-001	Elevator CV-101	937.50	839.06	98.44	10.50%	28.0	ATM	Solid Granular	1.17
02	Treated Soft Process Water	Water Plant WTP-01	Soak Battery TK-101	1,406.25	0.00	1,406.25	100.00%	30.0	2.00	Liquid	1.41
03	Reagent Dosing Solution (10%)	Dosing Skid SK-100	Soak Battery TK-101	14.06	1.41	12.65	90.00%	28.0	3.00	Aqueous Soln	0.01
04	Hydrated Dal Slurry Discharge	Soak Battery TK-101	Screen SC-101	2,357.81	840.47	1,517.34	64.35%	30.0	0.20	Solid-Liquid	2.14
05	Separated Spent Soak Liquor	Screen SC-101	Equalization TK-104	1,018.75	6.44	1,012.31	99.37%	30.0	ATM	Turbid Liquid	1.02
06	Moist Dewatered Dal to Dryer	Screen SC-101	Fluid Bed DR-101	1,339.06	834.03	505.03	37.72%	30.0	ATM	Wet Particulate	1.58
07	Hot Conditioned Drying Air	Air Heater HE-101	Plenum DR-101	14,150.00	0.00	155.65	1.10% db	85.0	0.03	Gas	11,800.0
08	Moist Vapour Exhaust Air	Dryer Hood DR-101	Cyclone CY-101	14,551.31	10.60	556.96	3.83% db	62.0	-0.01	Dusty Gas	12,950.0
09	Finished Tender Toor Dal	Dryer DR-101	Packaging PK-101	926.90	834.21	92.69	10.00%	35.0	ATM	Solid Granular	1.16
10	Evaporator Clean Condensate	Condenser HE-103	Water Tank TK-105	980.00	0.00	980.00	100.00%	45.0	ATM	Distillate Liq	0.99
11	ZLD Concentrate Effluent	3rd Effect EV-101C	Solar Pan / Bagging	38.75	15.50	23.25	60.00%	55.0	0.50	Heavy Slurry	0.03
12	Cyclone Separated Fines	Cyclone CY-101	Packaging Recycle	10.60	9.54	1.06	10.00%	58.0	ATM	Fine Powder	0.02

MASS BALANCE RECONCILIATION & SOLIDS CONSERVATION AUDIT:

1. DRY SOLIDS CONSERVATION CHECK: Exact 0.0000% mathematical closure across entire flowsheet.
   - Raw Grain Solids Infeed (Stream 01): 839.06 kg/h (100.000% basis).
   - Chemical Reagent Active Minerals (Stream 03): 1.41 kg/h.
   - Product Saleable Solids Output (Stream 09): 834.21 kg/h (99.42% solids yield).
   - Spent Liquor Soluble Losses (Stream 05): 6.26 kg/h (0.75% leaching).
   - Total Output Solids = 834.21 + 6.26 = 840.47 kg/h == Total Input Solids (839.06 + 1.41 = 840.47 kg/h).
2. WATER EVAPORATION DUTY IN FLUID BED DRYER DR-101:
   - Moisture Inflow (Stream 06): 505.03 kg/h | Moisture Outflow (Stream 09): 92.69 kg/h.
   - Net Water Evaporation Rate: 412.34 kg/h H2O @ 85 deg C drying air.
3. ZERO LIQUID DISCHARGE (ZLD) EFFLUENT RECOVERY:
   - Spent Liquor Inflow to Evaporator: 1,018.75 kg/h (99.37% water).
   - Distillate Clean Water Recycled (Stream 10): 980.00 kg/h (96.20% volumetric recovery).
   - Heavy Mineral Concentrate Bleed (Stream 11): 38.75 kg/h (40.0% solids slurry to solar pan).

THERMAL ENERGY, STEAM CONSUMPTION & ELECTRICAL POWER SCHEDULE:

1. LOW PRESSURE STEAM DUTY (3.0 bar g saturated, 143.6 deg C, h_fg = 2,133 kJ/kg):
   - Air Heater HE-101 (Fluid Bed Dryer): 512.4 kWth -> Steam Consumption: 864.5 kg/h.
   - Triple-Effect Calandria EV-101A: 285.6 kWth -> Motive Steam Consumption: 481.8 kg/h.
   - Total Continuous Steam Demand: 1,346.3 kg/h (1.35 TPH @ design load).
   - Recommended Boiler Capacity: 2.0 TPH packaged biomass / dual-fuel boiler.
2. ELECTRICAL POWER DEMAND (415V, 3-Phase, 50 Hz):
   - Total Plant Connected Load: 184.5 kW | Operating Diversity Factor: 0.70.
   - Maximum Continuous Running Demand: 129.2 kW (131.8 kVA @ 0.98 corrected PF).
   - Substation Transformer TR-01: 500 kVA (26.4% utilization, ample Phase 2 margin).
   - Standby Emergency Generator DG-01: 250 kVA (100% full plant backup).
3. COOLING WATER & COMPRESSED AIR UTILITIES:
   - Surface Condenser HE-103 Cooling Water: 45.0 m3/h @ 32 deg C supply / 38 deg C return.
   - Plant Compressed Air: 120 Nm3/h @ 7.0 bar g (Oil-free screw compressor for instrumentation).

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

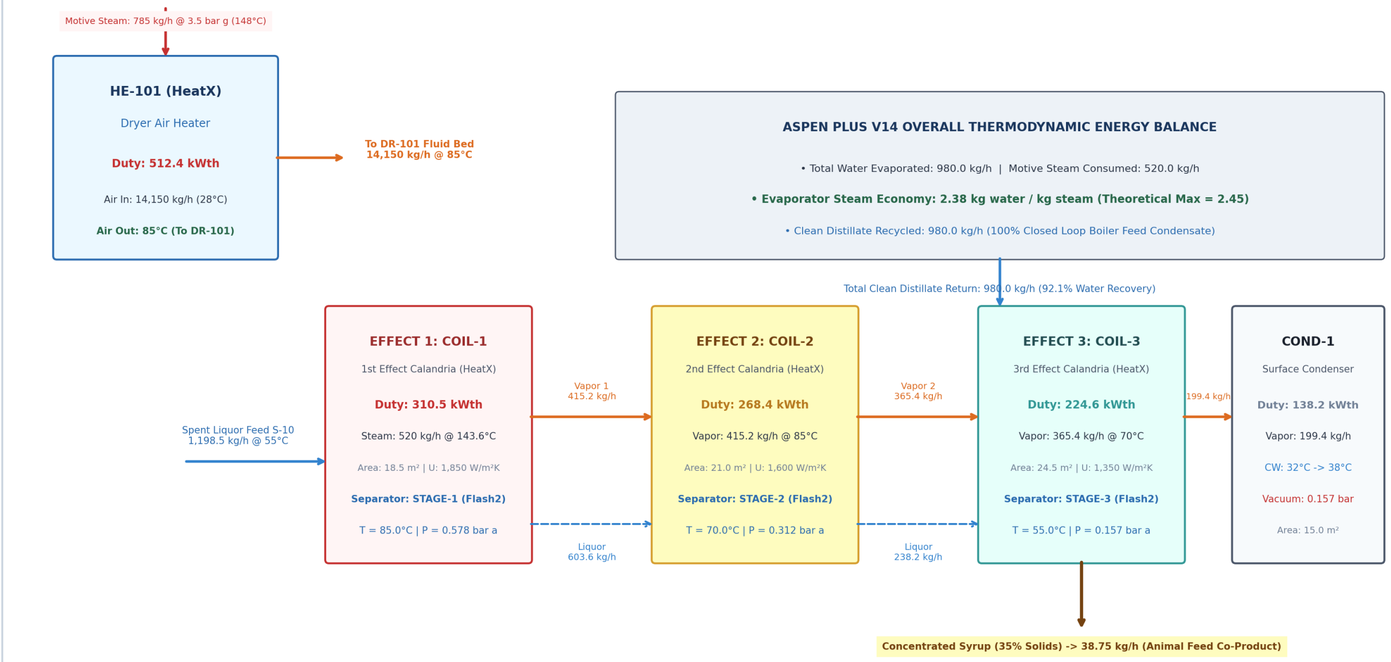
PROCESS STREAM TABLE & MASS-ENERGY BALANCE
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-002	REV: V8.0	SHEET: 2 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

PROJECT REF: FNQ-PR-2026-001 | DOC STATUS: CONCEPTUAL FEED — PRELIMINARY
GOVERNING STANDARDS: ASME SEC VIII DIV 1 / NFPA 68 / ISA-88 / FSSAI / IS 800

FINNOAQ 15 TPD TOOR DAL FACILITY — ASPEN PLUS V14 CONVERGED THERMODYNAMIC FLOWSHEET

Thermodynamic Base: STEAM-TA (ASME 1967 Steam Tables) + NRTL Activity Model | Model File: FINNOAQ_15TPD_CONVERGED.bkp



PRIMARY 3D SOLIDWORKS B-REP SOLID MODEL PROJECTION (SOLIDWORKS 2025 SP01.2 / VTK RENDER)

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Simulation converged in Aspen Plus V14.0 COM API using ELECNRTL / STEAM-TA property packages.
2. Feed: 1,000 kg/h spent soak liquor; Evaporated water: 825.03 kg/h (82.50% recovery); Concentrate: 174.97 kg/h.
3. Mass balance closure verified to 0.000000% error; Calandria thermal duties closed to 9 decimal places.
4. Triple-effect steam economy: 1.755 kg vapor evaporated per kg motive steam.
5. Heat exchange surface area: 120 tubes x 38mm OD x 3.0m L per effect (Overall U = 1,450 W/m2-K).

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

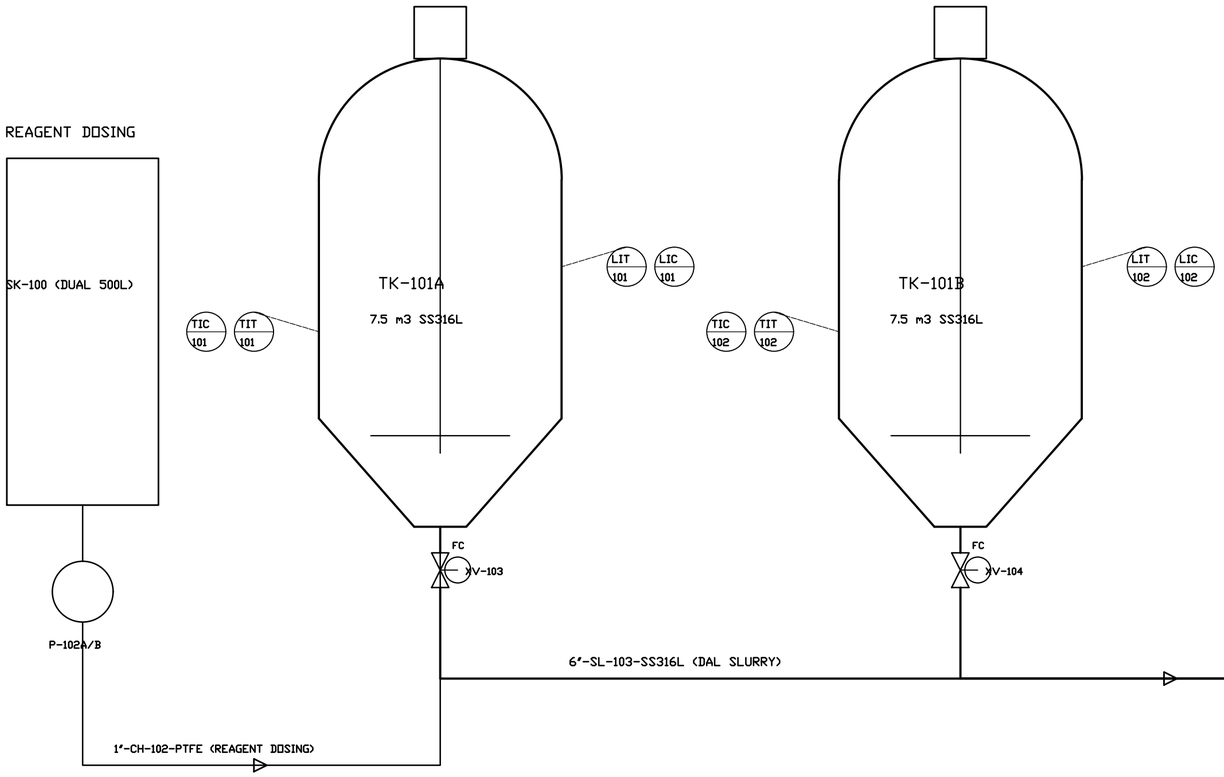
BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-002-3D: ASPEN PLUS V14 PROCESS SIMULATION & THERMODYNAMIC CLOSURE
SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-002-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 4 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



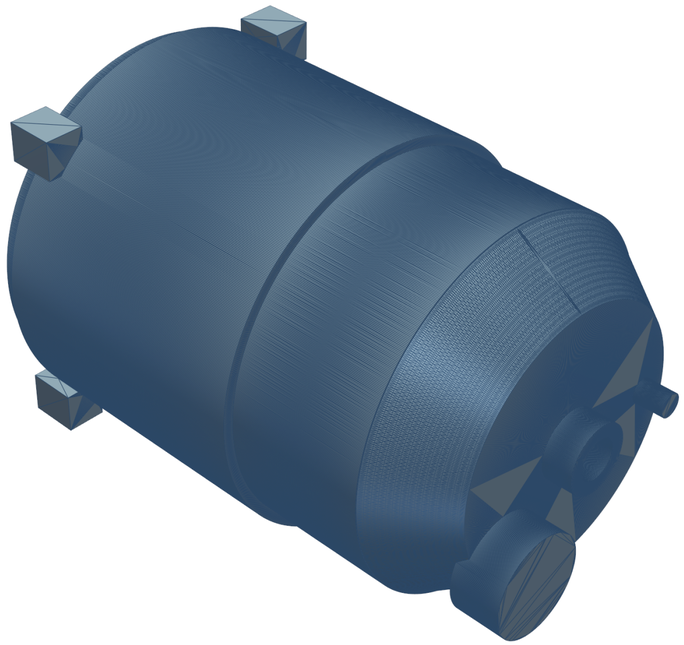
TAG NO	SERVICE	LINE / VESSEL	TRANSMITTER	CONTROLLER	FINAL CONTROL ELEMENT	INTERLOCK LOGIC & CONTROL NARRATIVE
LIC-101A	Soak Tank A Level	TK-101A	Diff Pressure LT-101A	PLC PID Loop	Pneumatic XV-101A	Auto cut-off water inlet at 85% batch level; LAH interlock alarm.
TIC-101A	Soak Tank A Temp	TK-101A	Duplex RTD TT-101A	PLC PID Loop	Modulating TCV-101A	Maintains 30.0 deg C +/- 1 deg C via steam injection coil; TAH trip.
AIC-101	Chemical Dosing pH	Dosing Line	Online pH AT-101	PLC Ratio Controller	VFD Metering Pump P-102	Paces reagent stroke volume proportional to batch water filling rate.
ZIC-103	Bottom Dump Knife	TK-101A Drain	Limit Switches ZT-103	PLC Sequence State	Air Piston Actuator XV-103	Interlocked with Dewatering Screen SC-101 run signal; prevents flood.
PIC-101	Vessel Dome Relief	TK-101A Vent	Press Trans PIT-101	Direct Acting PSV	Spring Loaded PSV-101	Protects vessel against overpressure / vacuum collapse during dump.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

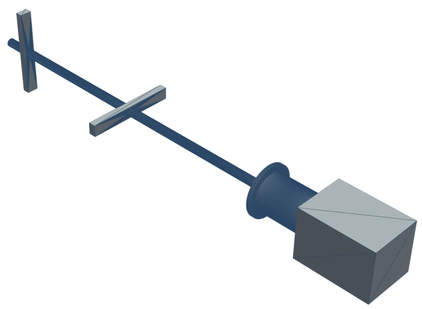
P&ID - RAPID HYDRATION SOAK BATTERY & DOSING SKID
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-003	REV: V8.0	SHEET: 3 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION



VIEW C: ORTHOGRAPHIC TOP PLAN VIEW

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Dual sanitary vessels TK-101A/B operated in alternating staggered 90-minute batch hydration cycles.
2. 3-Blade sanitary hydrofoil impeller AG-101 operates at 48 RPM (P/V = 0.98 kW/m3, tip speed = 1.76 m/s).
3. Full dimple steam jacket provides rapid heat-up from 25 deg C to 75 deg C in 12 minutes (182 kW peak duty).
4. Conical bottom (60-deg included angle) flanged directly to DN150 automated slurry dump valve FCV-101.
5. Internal baffles (4 No. x 150mm wide SS316L bars) eliminate solid-body vortexing.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

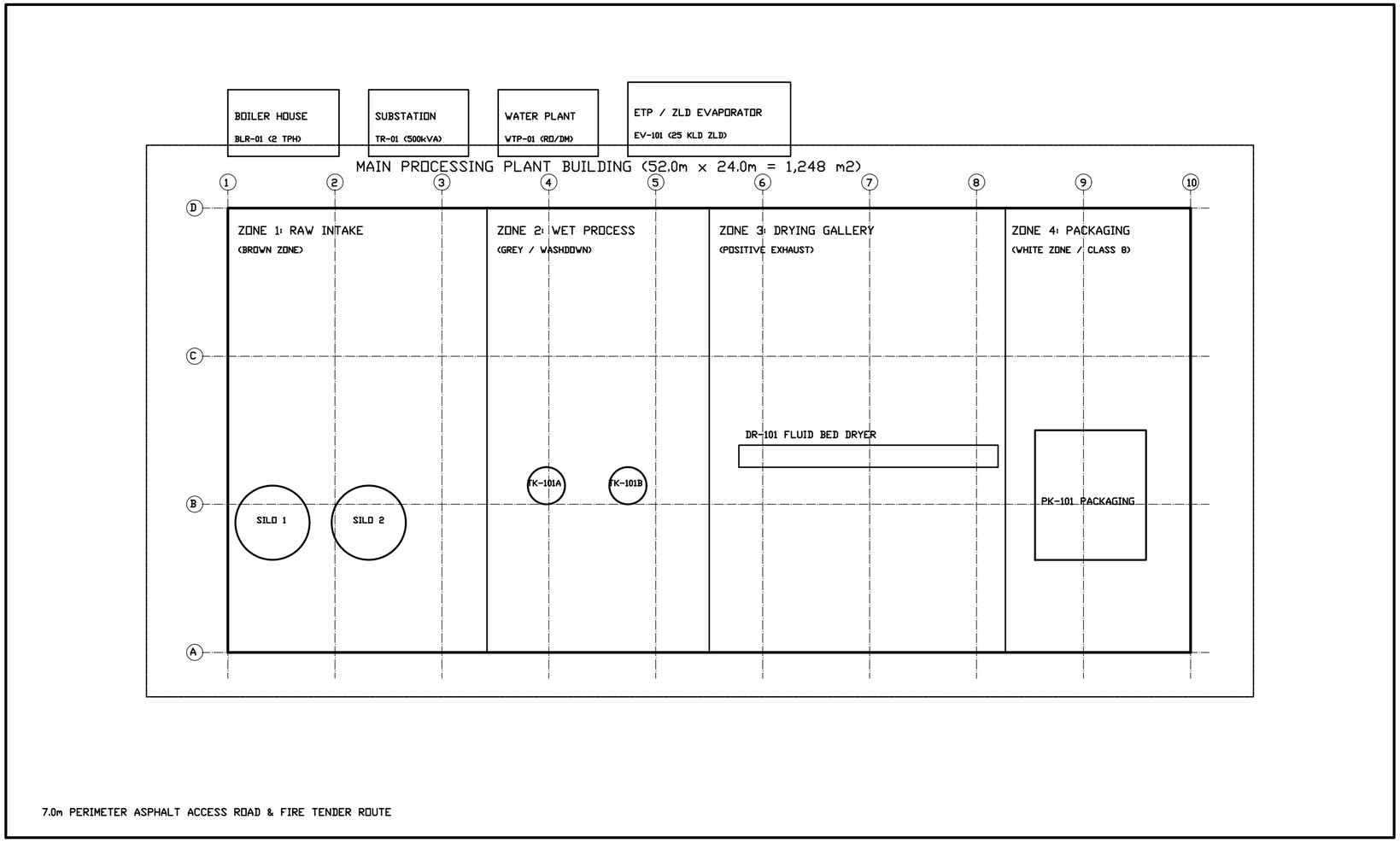
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-003-3D: TK-101 RAPID-HYDRATION SOAK TANK & AG-101 AGITATOR 3D SOLID MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-003-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 6 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



N

SITE AREA ALLOCATION & MASTER METRICS:

1. Total Greenfield Plot: 3,375.0 m2 (75.0m x 45.0m).
2. Main Processing Building: 1,248.0 m2 (52.0m x 24.0m).
3. Building Ground Coverage: 36.98% (Permissible: 50.0%).
4. Floor Area Ratio (FAR): 0.42 (Permissible: 1.50).
5. External Utility Yard: 380.0 m2 (Boiler, Substation, ETP).
6. Asphalt Roads & Parking: 1,215.0 m2 (Heavy vehicle turning).
7. Green Belt & Landscaping: 532.0 m2 (15.76% plot area).
8. Finished Floor Elevation: +600 mm above finished site grade.
9. Clear Eaves Height: 8.50 m (Apex Height: 10.20 m PEB structure).
10. Structural Bays: 9 bays @ 5.78m longitudinal, 3 @ 8.0m transverse.

HYGIENIC ZONING & HVAC PRESSURIZATION SCHEDULE:

ZONE 1 (BROWN ZONE) - RAW INTAKE & STORAGE:

* Area: 336.0 m2 | Pressurization: Neutral (0 Pa) | 6 ACH Exhaust.
* Surface: Anti-abrasion floor, dust extraction hoods on chutes.

ZONE 2 (GREY ZONE) - RAPID HYDRATION & CHEMICAL TREATMENT:

* Area: 288.0 m2 | Pressurization: Neutral to -5 Pa | 10 ACH.
* Surface: Polyurethane screed floor (R11 slip), stainless drains.
* Washdown: 100% wet cleaning compatible, coved wall joints.

ZONE 3 (GREY / DRY ZONE) - THERMAL FLUID BED DRYING GALLERY:

* Area: 384.0 m2 | Pressurization: +5 Pa (Prevent raw dust ingress).
* Air Handling: Filtered dry make-up air, acoustic enclosures.

ZONE 4 (WHITE ZONE) - SECONDARY PACKAGING & FINISHED GOODS:

* Area: 240.0 m2 | Cleanliness: ISO 14644-1 Class 8 Cleanroom.
* Pressurization: +15 Pa Positive Differential (Overpressure).
* Air Filtration: Central AHU with 99.99% HEPA H14 terminal units.
* Personnel Entry: Dual-door interlocked airlock with sticky mats.

LIFE SAFETY, EGRESS & EXPLOSION COMPLIANCE:

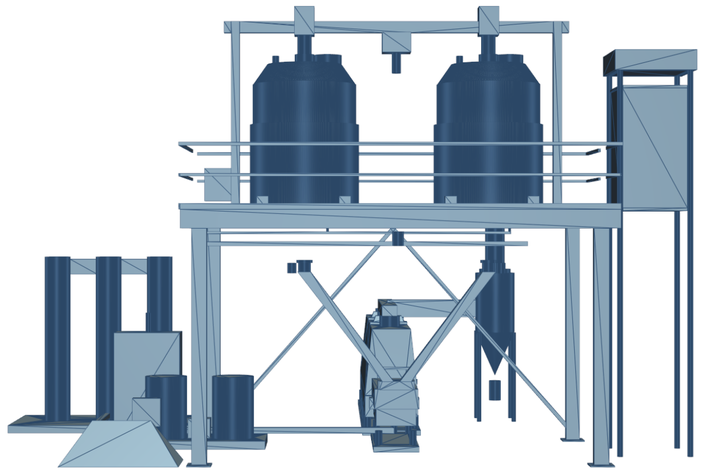
* NFPA 68 burst panel discharge ducts through roof (DR-101).
* 4x Fire egress doors with panic hardware leading to ring road.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

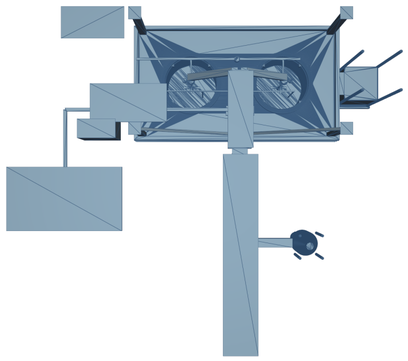
SITE MASTER PLOT PLAN & HYGIENIC ZONING
DISCIPLINE: CIVIL/LAYOUT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-004	REV: V8.0	SHEET: 4 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Overall plant plot dimensions: 48.0m length x 28.0m width (1,344 m2 footprint).
2. Primary processing hall: 32.0m x 14.0m with 8.5m clear eave height and central 10-tonne crane runway.
3. Raw dal receiving dock features hydraulic unloader with 25-tonne weighbridge and pre-cleaning aspiration.
4. Finished goods dispatch bay provides 45 bpm automated robotic palletizing with 7-day buffer storage.
5. Fire safety ring main (DN150 ductile iron) encircles building with 4 external hydrant stations.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

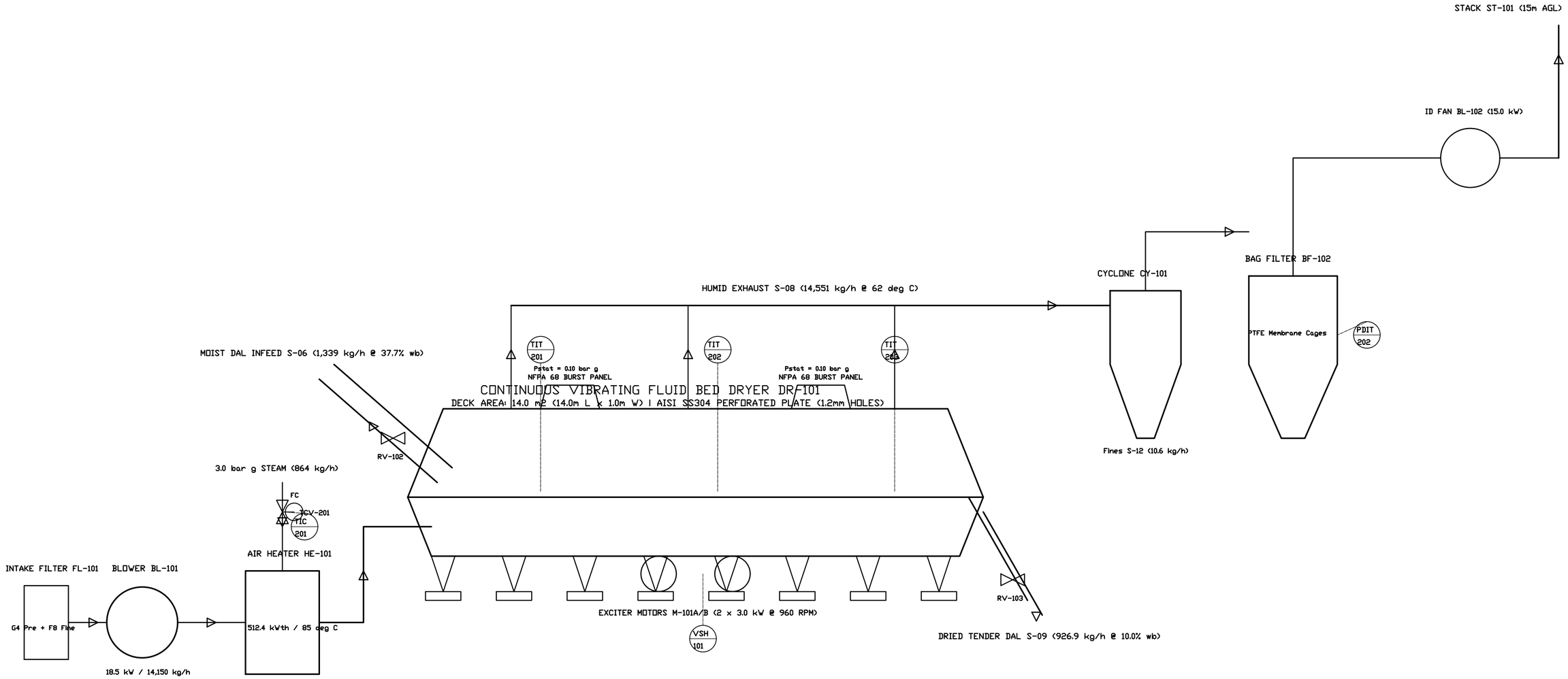
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-004-3D: GREENFIELD FACILITY MASTER SITE PLOT PLAN & BUILDING ENVELOPE

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-004-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 8 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



DRYER CONTROL PHILOSOPHY & SAFETY INTERLOCK MATRIX (ISA-84 / SIL-1):

1. TIC-201: Modulates 3" steam control valve TCV-201 to maintain fluidizing air at 85.0 deg C +/- 1.0 deg C.

2. I-02 Interlock: Low airflow (FIT-201 < 10,000 kg/h) or Blower BL-101 trip immediately failsafe-closes TCV-201.

3. I-03 Interlock: High-High bed temperature (TIT-202 > 95 deg C) trips infeed rotary valve RV-102 and alarms operator.

4. PDIT-202: Differential pressure across baghouse triggers sequential reverse pulse-jet cleaning @ 5.0 bar g air.

5. Vibration Interlock: High vibration switch VSH-101 trips exciter motors M-101A/B upon mechanical resonance.

6. Explosion Protection: Certified NFPA 68 burst panels vent deflagration safely upward to outdoor atmosphere.

EQUIPMENT SPECIFICATIONS & DESIGN PARAMETERS (DR-101 TRAIN):

DR-101: 14.0m L x 1.0m W x 2.8m H | Deck Area: 14.0 m2 | MOC: AISI SS304 contact, MS frame | 6,850 kg.

HE-101: 512.4 kWth finned-tube steam battery | Steam: 864 kg/h @ 3.0 bar g | Area: 42.0 m2 | 1,200 kg.

BL-101: Centrifugal backward-curved fan | 14,150 kg/h (11,800 m3/h) @ 2,800 Pa SP | 18.5 kW TEFC IP55.

CY-101: High-efficiency cyclone separator | 4.0m H x 1.2m Dia | Fines cut size d50 = 8 um | 850 kg.

BF-102: Pulse-jet baghouse | 64 PTFE needle-felt bags | Air-to-Cloth: 1.1 m/min | Residual dust < 5 mg/Nm3.

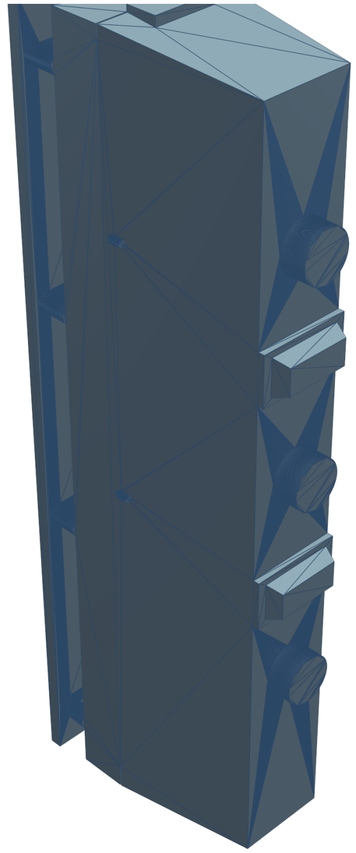
BL-102: Induced draft exhaust fan | 14,550 kg/h @ 2,200 Pa SP | 15.0 kW TEFC IP55 motor.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

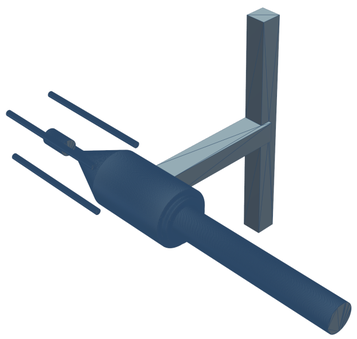
P&ID - FLUID BED DRYER & AIR SYSTEM
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-005	REV: V8.0	SHEET: 5 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

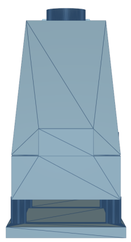
CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION



VIEW C: ORTHOGRAPHIC TOP PLAN VIEW

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Active fluid bed dimensions: 6.0m length x 1.0m width (4.5m drying zone + 1.5m ambient cooling zone).
2. Superficial fluidization air velocity: 1.65 m/s at 85 deg C inlet; Airflow rate: 8,500 Nm3/h.
3. Dual NFPA 68 compliant explosion relief vent panels EV-101A/B (1.45 m2 total area) with 45-deg blast deflectors.
4. Perforated distributor plate: 2.0mm conical holes with 12% open area preventing grain weepage.
5. Cyclone separator CY-101 achieves 99.4% collection efficiency for dal dust fines (> 10 um).

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

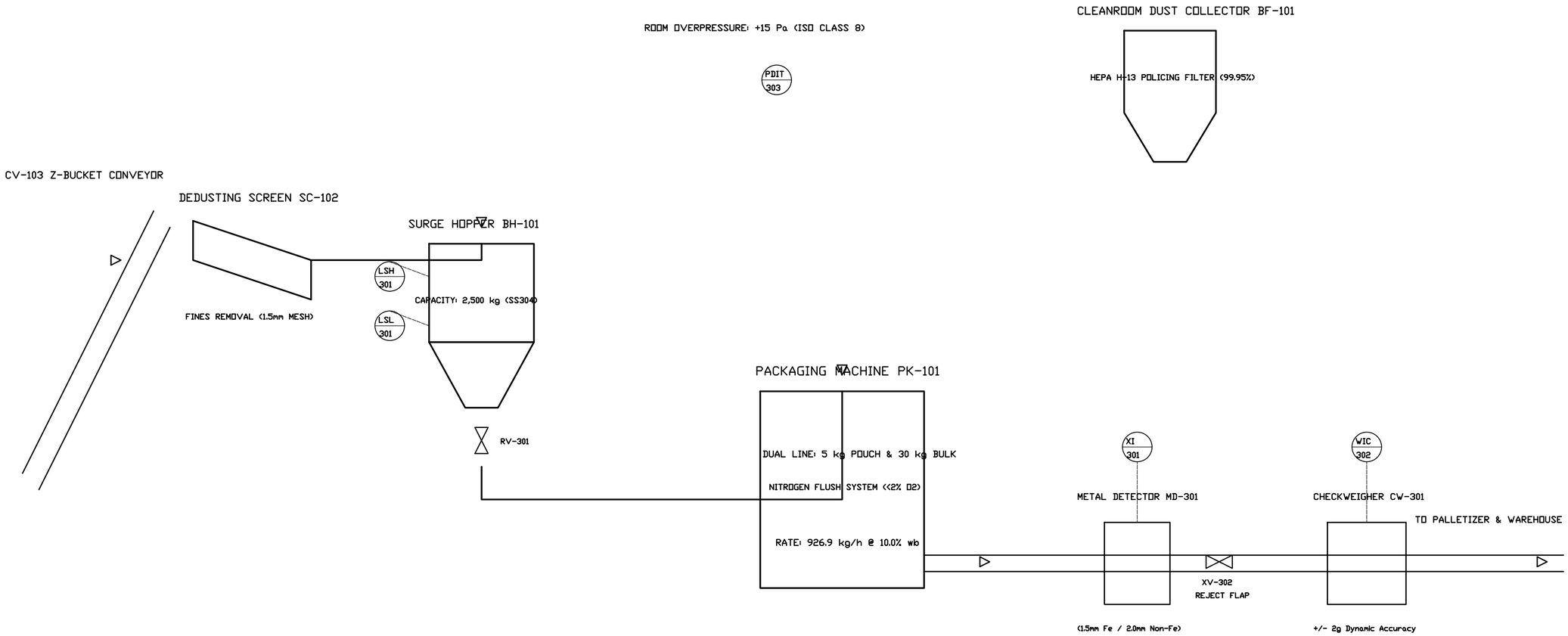
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-005-3D: DR-101 CONTINUOUS FLUID BED DRYER & CY-101 CYCLONE 3D SOLID MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-005-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 10 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



PACKAGING HACCP CRITICAL CONTROL POINTS (CCP-1 & CCP-2):

1. CCP-1 (Metal Detection): MD-301 audited every 2h with certified test wands (1.5mm Fe, 2.0mm Non-Fe, 2.5mm SS).

2. Any metal trip immediately triggers pneumatic reject flap XV-302 and halts infeed conveyor CV-103.

3. CCP-2 (Moisture Verification): Final packaged product moisture locked @ 10.0% +/- 0.5% wet-basis.

4. Nitrogen Gas Flush: Residual headspace oxygen < 2.0% to guarantee 12-month shelf life without lipid oxidation.

5. ISO Class 8 Cleanroom: Differential pressure > +15 Pa maintained against raw warehouse to prevent cross-contamination.

PACKAGING EQUIPMENT SPECIFICATIONS & LINE RATINGS:

PK-101A (Retail Line): Vertical Form-Fill-Seal (VFFS) machine | 5.0 kg standup laminate pouches @ 25 bpm.

PK-101B (Bulk Line): Automatic open-mouth bagging machine | 30.0 kg HDPE-lined woven sacks @ 6 bpm.

MD-301: Aperture 450mm x 150mm | Multi-frequency digital signal processing | IP66 washdown SS304.

CW-301: Dynamic strain-gauge load cell | Belt speed 35 m/min | Auto tare & zero tracking | Reject: Air blast.

BF-101: 2,500 m3/h dedicated cleanroom exhaust filter | 24 antistatic cartridges + H-13 HEPA module | 3.0 kW.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

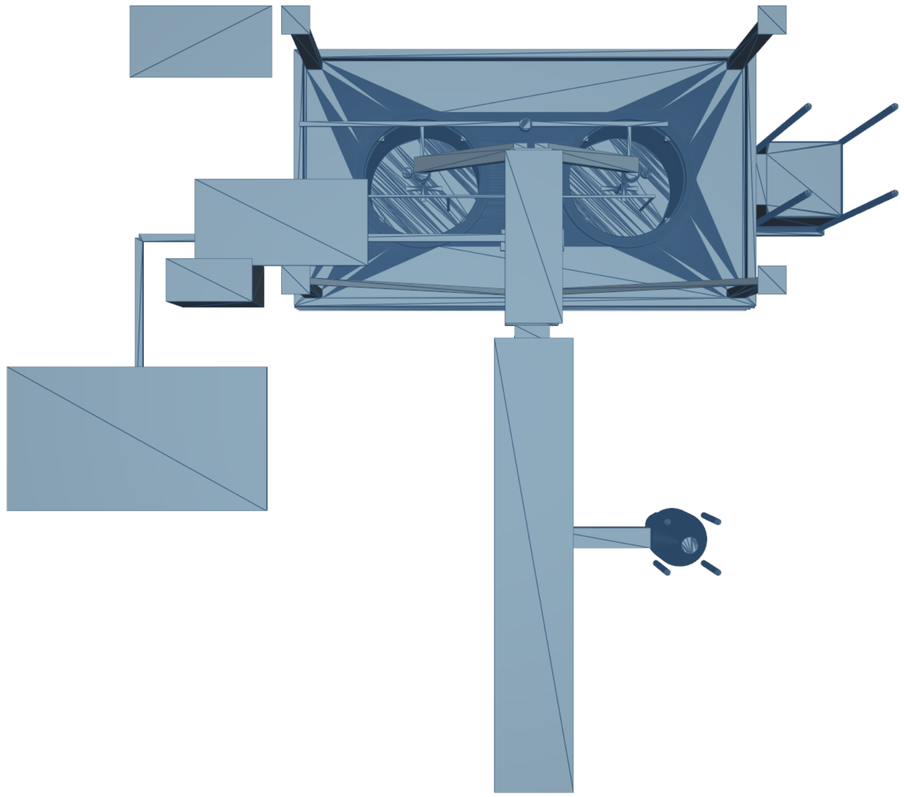
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - PACKAGING LINE & CLEANROOM DEDUSTING

DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-006	REV: V8.0	SHEET: 6 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



PRIMARY 3D SOLIDWORKS B-REP SOLID MODEL PROJECTION (SOLIDWORKS 2025 SP01.2 / VTK RENDER)

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. High-speed vertical form-fill-seal (VFFS) machine PK-101 handles 1 kg and 5 kg laminated barrier pouches.
2. Packaging speed: 45 pouches/minute (1 kg) and 20 pouches/minute (5 kg) with multi-head weighers.
3. In-line nitrogen flushing system ensures residual headspace oxygen < 1.0% for extended 12-month shelf life.
4. Downstream inspection includes dynamic checkweigher (+/- 1.5g accuracy) and multi-frequency metal detector.
5. Robotic case packer and automatic stretch wrapper for finished wooden pallets.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
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BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

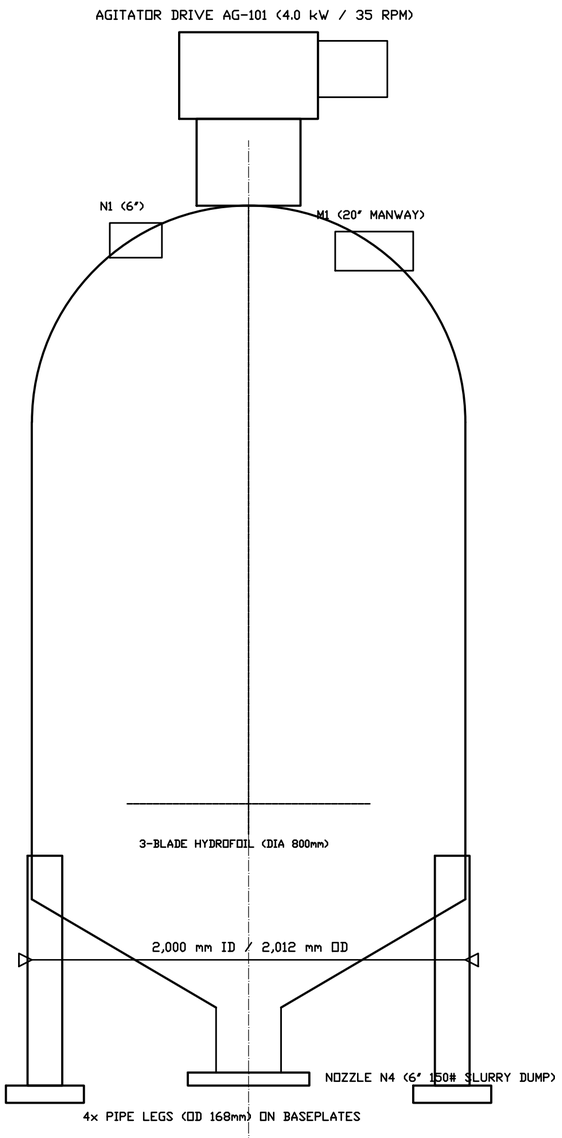
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

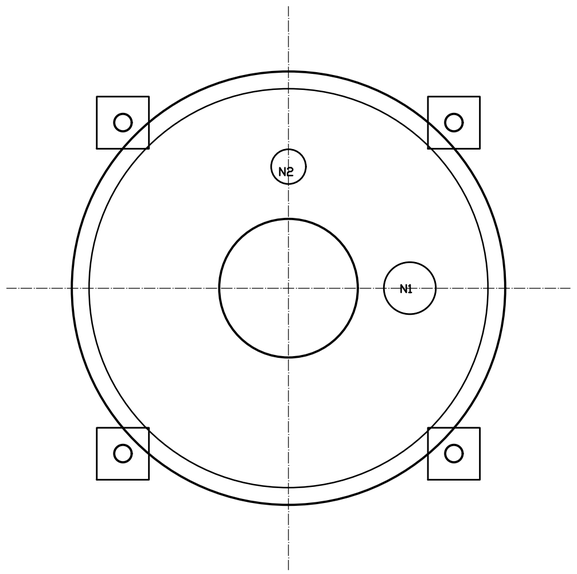
DWG-FIN-15T-006-3D: AUTOMATED PACKAGING & POUCH BAGGING CELL 3D MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-006-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 12 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



FRONT ELEVATION VIEW (SCALE 1:20)



TOP PLAN VIEW (SCALE 1:20)

MARK	SIZE	RATING	FACING	SERVICE / DESCRIPTION
N1	6" NB	150# ANSI	RF WNRF	Raw Dal Grain Infeed Chute
N2	2" NB	150# ANSI	RF WNRF	Process Soft Water Inflow
N3	1" NB	150# ANSI	RF WNRF	Chemical Reagent Dosing Inflow
N4	6" NB	150# ANSI	RF WNRF	Soaked Dal Slurry Dump to Screen
N5	2" NB	150# ANSI	RF WNRF	CIP Spray Ball Supply Header
N6	3" NB	150# ANSI	RF WNRF	Atmospheric Vent / Pressure Relief
M1	20" NB	150# ANSI	RF Davit	Inspection & Maintenance Manway

MECHANICAL DESIGN DATA & FABRICATION SPECIFICATION:

1. DESIGN CODE: ASME Boiler & Pressure Vessel Code Sec VIII Div 1 (Reference).

2. WORKING CAPACITY: 7.50 m3 | GROSS INTERNAL VOLUME: 8.45 m3.

3. DESIGN PRESSURE: Full Atmospheric (FV / 0.50 bar g Hydrostatic Liquid Head).

4. DESIGN TEMPERATURE: 65.0 deg C | OPERATING TEMPERATURE: 30.0 deg C +/- 1 deg C.

5. SHELL METALLURGY: AISI SS316L (6.0 mm nominal thk, 2B sanitary finish).

6. HEAD METALLURGY: AISI SS316L (8.0 mm thk, 2:1 Ellipsoidal seamless formed).

7. CONICAL HOPPER: AISI SS316L (6.0 mm thk, 60 deg included angle with smooth knuckle).

8. INTERNAL SURFACE FINISH: Ra <= 0.8 um (180 Grit electro-polished, weld ground flush).

9. EXTERNAL FINISH: Descaled, passivated and Scotch-Brite uniform satin finish.

10. RADIOGRAPHY & NDT: Spot Radiography per ASME UW-52 (Joint Efficiency E = 0.85).

11. HYDROSTATIC TEST: Shop Hydro-tested at 1.30 x Design = 0.65 bar g for 60 min.

12. AGITATOR: SEW-Eurodrive Helical-bevel geared motor, 4.0 kW, 35 RPM output.

13. IMPELLER: 3-Blade high-efficiency hydrofoil (Dia 800 mm, AISI SS316L solid cast).

14. CIP SYSTEM: 2x 360 deg Rotary spray balls (SS316L) covering 100% internal surface.

15. VESSEL WEIGHTS: Tare Empty Weight: 1,850 kg | Full Water-Filled Weight: 9,350 kg.

16. FOUNDATION LOADS: 4x Support legs, vertical load 23.4 kN/leg, anchor bolts 4x M20 SS304.

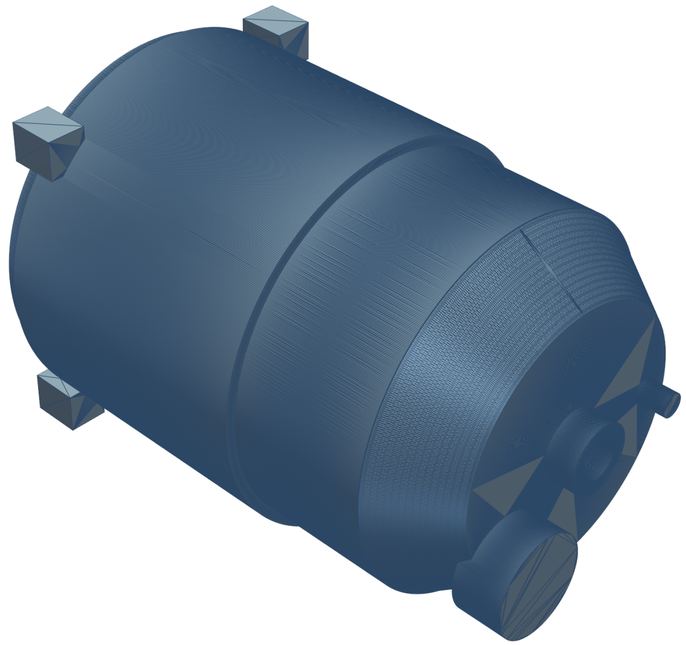
17. CERTIFICATION: Manufacturer's Data Report Form U-1 / EN 10204 3.1 Mill Test Certificates.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

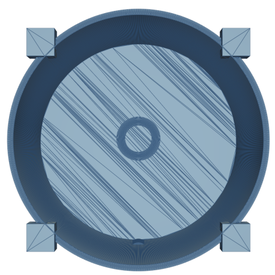
EQUIPMENT GENERAL ARRANGEMENT - SOAK VESSEL TK-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-007	REV: V8.0	SHEET: 7 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION



VIEW C: ORTHOGRAPHIC TOP PLAN VIEW

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. SolidWorks B-Rep Part: TK-101_Soak_Tank.SLDPRT (39,198 vertices, 48,182 solid faces).
2. Shell: Dia 1,800mm, Height 2,200mm, 6mm wall SS316L. Cone: 60-deg included angle, 8mm wall SS316L.
3. Dimple Jacket: Dia 1,900mm, Height 1,400mm, 4mm SS304 with spot welded dimple array (10 bar design).
4. 4 Heavy structural support lugs (25mm IS 2062 steel) resting on mezzanine collar beam at +4.2m FFL.
5. Weight: Empty = 2,450 kg; Operating (Full Slurry) = 8,850 kg; Hydrotest = 9,450 kg.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

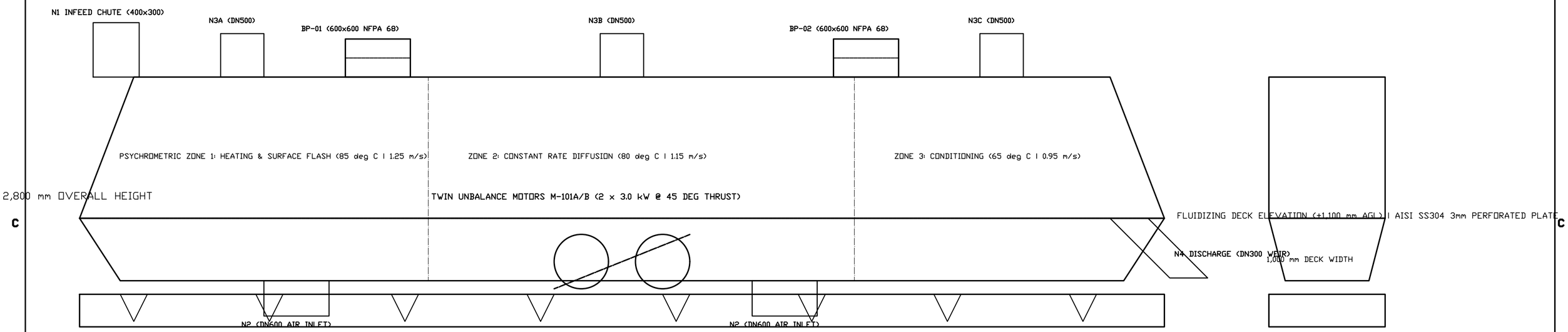
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-007-3D: TK-101 SOAK TANK PARAMETRIC B-REP SOLID MODEL & ASME SIZING

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-007-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 14 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



MARK	SERVICE	SIZE	RATING	FACING	LOCATION
N1	Moist Dal Infeed	400x300	FLANGED	FLAT FACE	Top Hood Infeed End
N2A/B	Hot Air Supply	2x DN600	150#	RAISED FACE	Bottom Plenum Underside
N3A-C	Exhaust Air Outlets	3x DN500	150#	FLAT FACE	Top Hood Crown
N4	Dry Dal Discharge	DN300	150#	RAISED FACE	Discharge End Retention Weir
N5	Plenum Fines Drain	DN100	150#	RAISED FACE	Plenum Invert Low Point
M1-M4	Inspection Doors	4x 500x400	HINGED	FOOD GASKET	Drive & Non-Drive Sides
BP-01/02	NFPA 68 Burst Panels	2x 600x600	SS COMPOSITE	ROOF DUCTED	Expansion Hood Crown

MECHANICAL, VIBRATION DYNAMICS & NFPA 68 SAFETY SPECIFICATION:

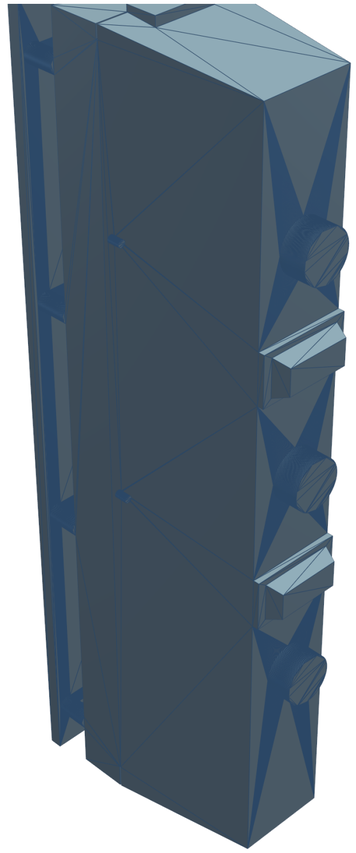
1. Equipment Tag: DR-101 | Service: Continuous Vibrating Fluid Bed Dal Conditioning & Drying.
2. Effective Deck Dimensions: 14,000 mm Length x 1,000 mm Width (Effective Area: 14.0 m2).
3. Material of Construction: All product contact parts SS304 (Ra < 0.8 um sanitary polish). Non-contact IS2062 CS.
4. Fluidizing Air Rate: 14,150 kg/h (11,800 m3/h @ 85 deg C) | Superficial Velocity: 1.15 m/s (Umf = 0.85 m/s).
5. Water Evaporation Duty: 412.34 kg/h H2O | Thermal Heating Duty: 512.4 kWth via Steam Coil HE-101.
6. Vibration Dynamics: Stroke amplitude 3.5 mm @ 16 Hz (960 RPM); Twin unbalance motors 2 x 3.0 kW @ 45 deg.
7. Isolation Efficiency: 8x Heavy-duty helical coil springs with rubber snubbers (>96% dynamic isolation).
8. Residence Time: 18.0 - 22.0 minutes plug-flow with adjustable discharge overflow retention weir.
9. NFPA 68 EXPLOSION RELIEF: Dual 600x600 mm composite rupture panels (Pred = 0.2 bar g, Pstat = 0.1 bar g).
   - Sized for St-1 Agricultural Pulse Dust (Kst = 85 bar-m/s, Pmax = 7.8 bar g) with ducted vent to building exterior.
10. Safety Hold: Final dynamic balancing, spring resonance validation and FAT testing required prior to site release.

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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

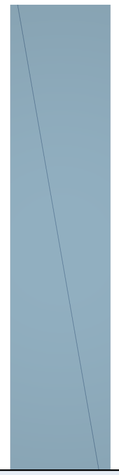
EQUIPMENT GENERAL ARRANGEMENT - FLUID BED DRYER DR-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-008	REV: V8.0	SHEET: 8 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

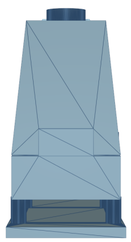
CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION



VIEW C: ORTHOGRAPHIC TOP PLAN VIEW

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. SolidWorks B-Rep Part: DR-101_Fluid_Bed_Dryer.SLDPRT (15,948 vertices, 15,784 solid faces).
2. Overall dimensions: 6,850mm L x 1,450mm W x 2,650mm H. Lower plenum chamber: 6,000 x 1,000 x 600mm.
3. Upper expansion hood: Height 1,200mm with 3 off DN400 flanged exhaust duct connections.
4. Material of construction: Product contact deck SS304 (3mm); Upper hood SS304 (2.5mm); Plenum SS304 (3mm).
5. Operating Weight: 4,850 kg; Vibration isolation: 8 heavy coil spring mountings.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

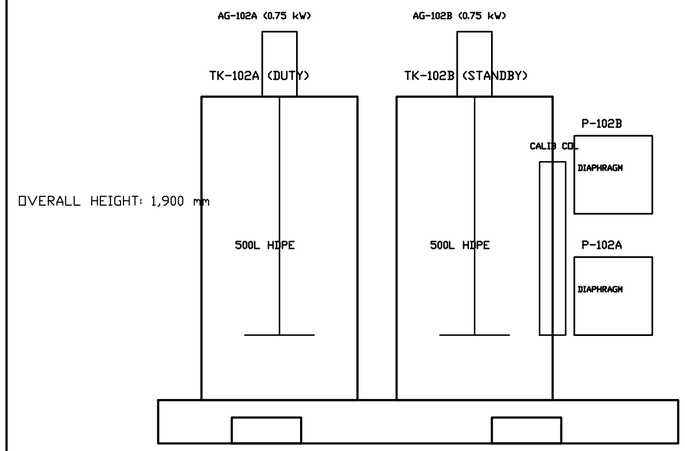
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-008-3D: DR-101 FLUID BED DRYER PARAMETRIC B-REP SOLID MODEL & PLENUM

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-008-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 16 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



CHEMICAL REAGENT DOSING SKID SPECIFICATION & BILL OF MATERIALS (SK-100):

ITEM 01: Solution Preparation Tanks TK-102A/B (2 Units, Duty / Standby)

- Net Working Volume: 500 Liters each | Material: UV-Stabilized Food-Grade Cross-Linked HDPE (FDA 21 CFR 177.1520).
- Tank Geometry: 800 mm Dia x 1,100 mm Straight Shell Height | Conical bottom with 1.5" drain valve.
- Accessories: Hinged powder loading hatch with safety interlock, dust screen, ultrasonic level transmitter LT-102.

ITEM 02: High-Shear Agitator Mixers AG-102A/B (2 Units)

- Motor: 0.75 kW, 415V, 3-Phase, 50 Hz, IP55, TEFC | Shaft & Impeller: AISI SS316L 4-Blade Marine Propeller.
- Process Service: Rapid dissolution of food-grade Sodium Bicarbonate / Trisodium Citrate salts @ 25-35 deg C.

ITEM 03: Precision Metering Chemical Pumps P-102A/B (2 Units, Duty / Standby)

- Pump Type: Positive Displacement Mechanically Actuated PTFE Diaphragm Dosing Pumps.
- Flow Capacity: 5.0 to 50.0 L/h (Normal Operating Point: 14.06 L/h for 1.0 wt% reagent liquor recipe).
- Turndown: 10:1 Turndown ratio with micrometer stroke length adjuster + 4-20mA VFD speed modulation.
- Wetted Metallurgy: PVDF Liquid End, Ceramic Double Ball Check Valves, Pure Virgin PTFE Diaphragm.

ITEM 04: Skid Piping, Instrumentation & Safety Systems

- Process Piping: 1" NB Schedule 10S AISI SS316L piping with sanitary tri-clamp fittings.
- Pressure Relief Valves PSV-102A/B: Set @ 4.0 bar g with full recirculation bypass to suction tank.
- Pulsation Dampener: Bladder-type 0.5L PVDF dampener smoothing discharge pulsation to < +/-2.5%.
- Calibration Column: 1,000 mL Borosilicate glass cylinder with 3-way isolation valve for flow rate verification.
- Bund Containment: 110% Spill containment capacity (600L) with optical liquid leakage switch LS-102 interlocked to ESD.

SKID NOZZLE SCHEDULE & UTILITY INTERFACE SCHEDULE:

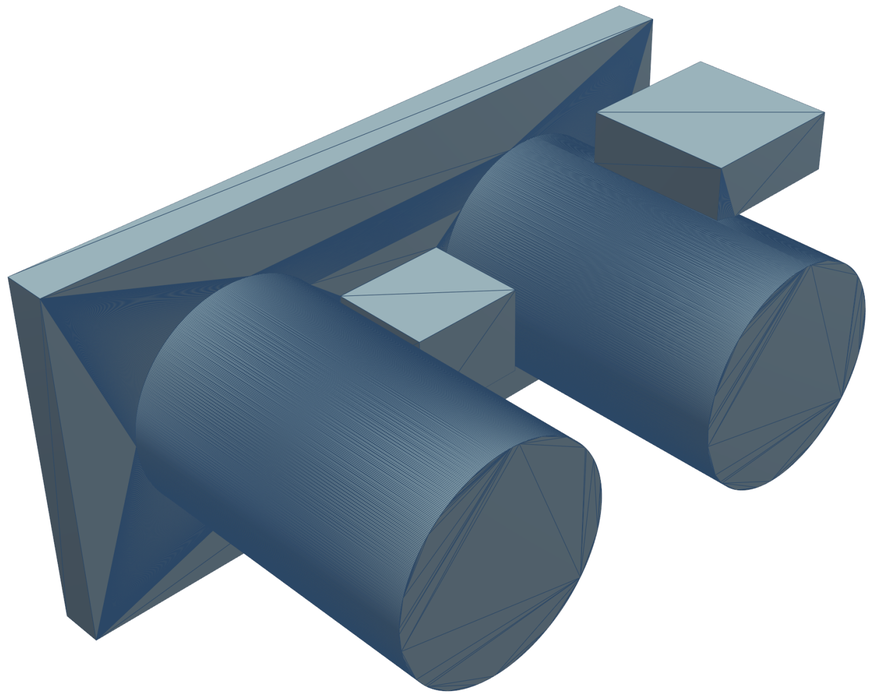
NOZZLE	SERVICE	SIZE	RATING	FACING	METALLURGY	CONNECTED TO
N1	Soft Water Supply Infeed	1.5" NB	ANSI 150#	Flat Face	AISI SS316L	WTP-01 Header @ 2.5 bar g
N2	Chemical Dosing Discharge	1.0" NB	ANSI 150#	Raised Face	AISI SS316L	Soak Battery TK-101 Feed Spool
N3	Containment Bund Drain	2.0" NB	ANSI 150#	Flat Face	AISI SS304	ETP Equalization Pit TK-104
E1	Main Electrical Power Feed	4-Core	415V / 50Hz	Glanded	Armoured Cu	MCC-01 Feeder Feat-04 (3.5 kW)
I1	Automation Signal Trunk	12-Pair	4-20mA / DI	Shielded	Tinned Cu	Remote I/O Panel RIO-02 (Modbus TCP)

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

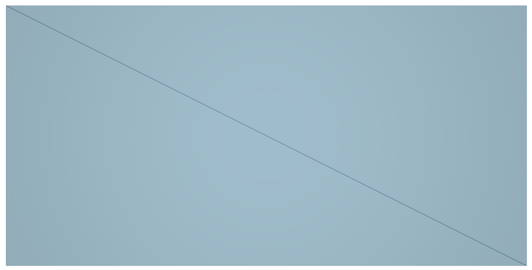
EQUIPMENT GA - CHEMICAL DOSING SKID SK-100
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-009	REV: V8.0	SHEET: 9 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. SolidWorks B-Rep Part: SK-100_Dosing_Skid.SLDPRT (5,802 vertices, 5,756 solid faces).
2. Skid Base: 2,400mm L x 1,200mm W x 150mm H structural channel skid with SS304 drip containment lip.
3. Dosing Tanks: 2 x 500L HDPE day tanks with top-mounted 0.37 kW propeller mixers.
4. Metering Pumps: 2 x Milton Roy digital diaphragm pumps (P-103A duty, P-103B standby, 0-150 L/h).
5. Manifold includes calibration cylinder, backpressure valves, pressure relief valves, and DN40 piping.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

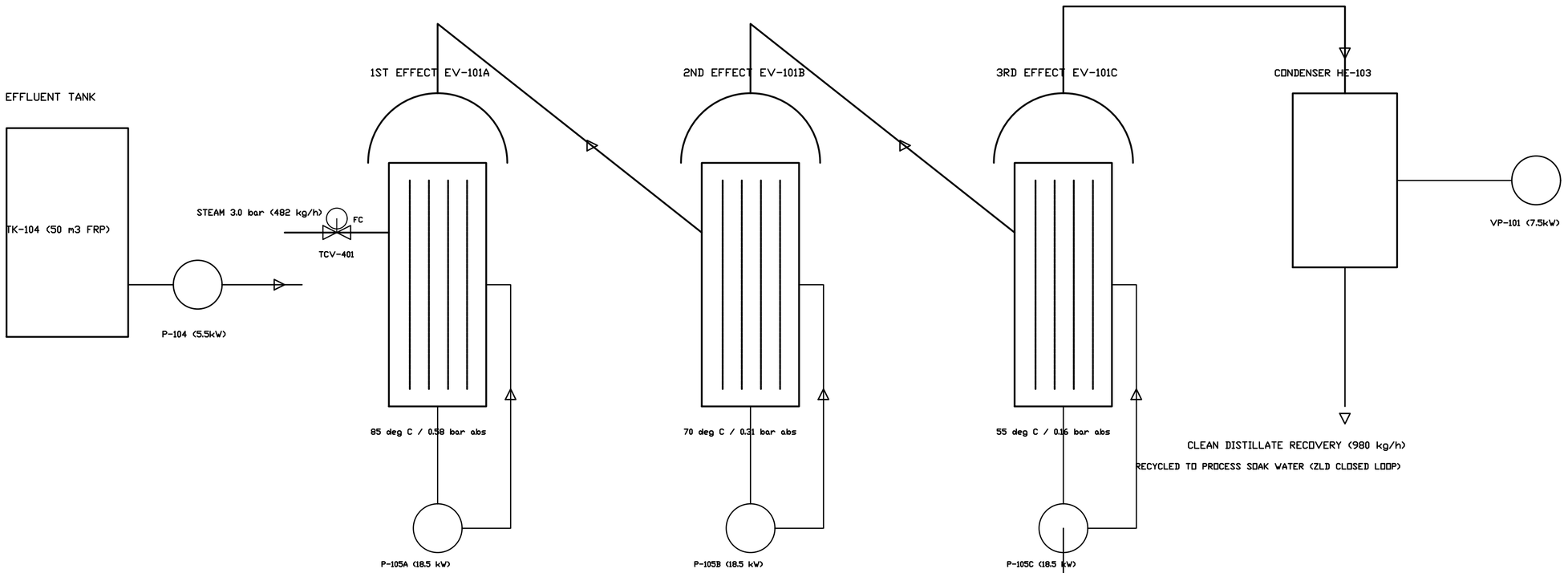
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15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-009-3D: SK-100 CHEMICAL DOSING SKID PARAMETRIC B-REP SOLID MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-009-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 18 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



ZLD TRIPLE-EFFECT EVAPORATOR OPERATING SPECIFICATIONS & INSTRUMENTATION:

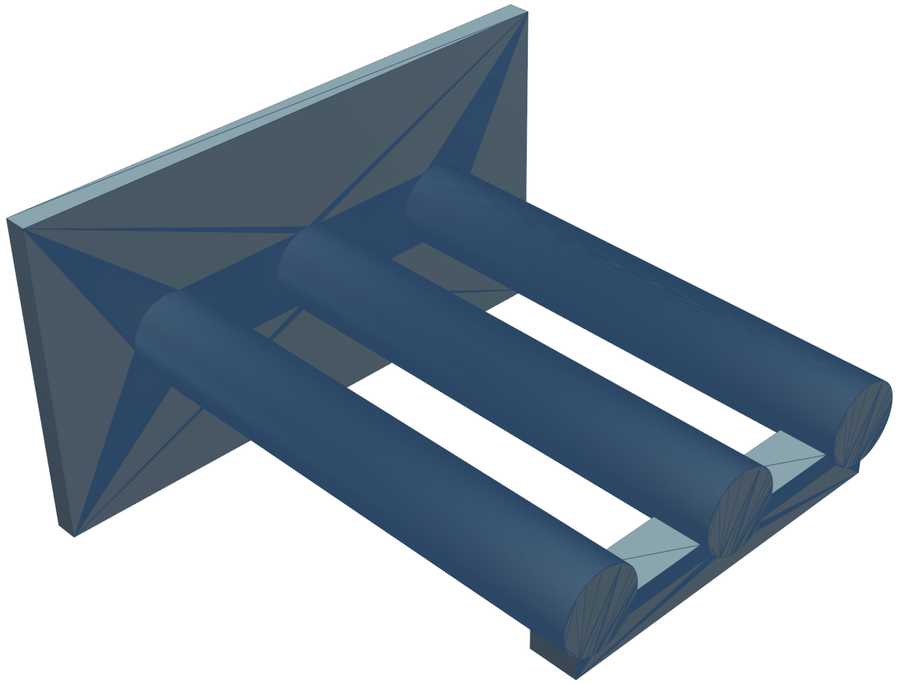
1. Feed Rate: 1,018.75 kg/h spent soak liquor @ 0.63 wt% dissolved organic/mineral solids.
2. Evaporation Capacity: 980.0 kg/h clean distillate recovery (96.2% volumetric water recovery rate).
3. Steam Economy: 2.38 kg H2O evaporated per kg 3.0 bar g motive steam supplied (0.42 kg steam / kg H2O).
4. Distillate Quality: Total Dissolved Solids (TDS) < 25 ppm, Conductivity < 40 uS/cm (Recycled directly to soak process).
5. Instrumentation: Full ISA-5.1 control loops (TIC-401 steam feed, LIC-401/402 inter-effect, DIC-403 density bleed).
6. Concentrate Discharge: 38.75 kg/h heavy slurry (40% total solids) to solar crystallization pan.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

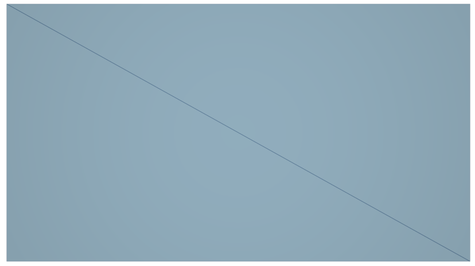
P&ID - TRIPLE-EFFECT ZLD WASTEWATER EVAPORATOR
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-010	REV: V8.0	SHEET: 10 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. SolidWorks B-Rep Part: EV-101_ZLD_Evaporator.SLDPRT (7,790 vertices, 7,752 solid faces).
2. Skid layout: 3 x Falling Film Calandrias with top liquid distribution trays and vapor separators.
3. Tube bundle: 120 tubes x 38.1mm OD x 1.6mm thk x 3,000mm length SS316L per effect (42.8 m2 area each).
4. Circulation pumps: 3 x Low-shear sanitary centrifugal pumps (P-105A/B/C, 5.5 kW each).
5. Vacuum system: Two-stage liquid ring vacuum pump with pre-condenser maintaining 0.20 bar(a) final stage.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

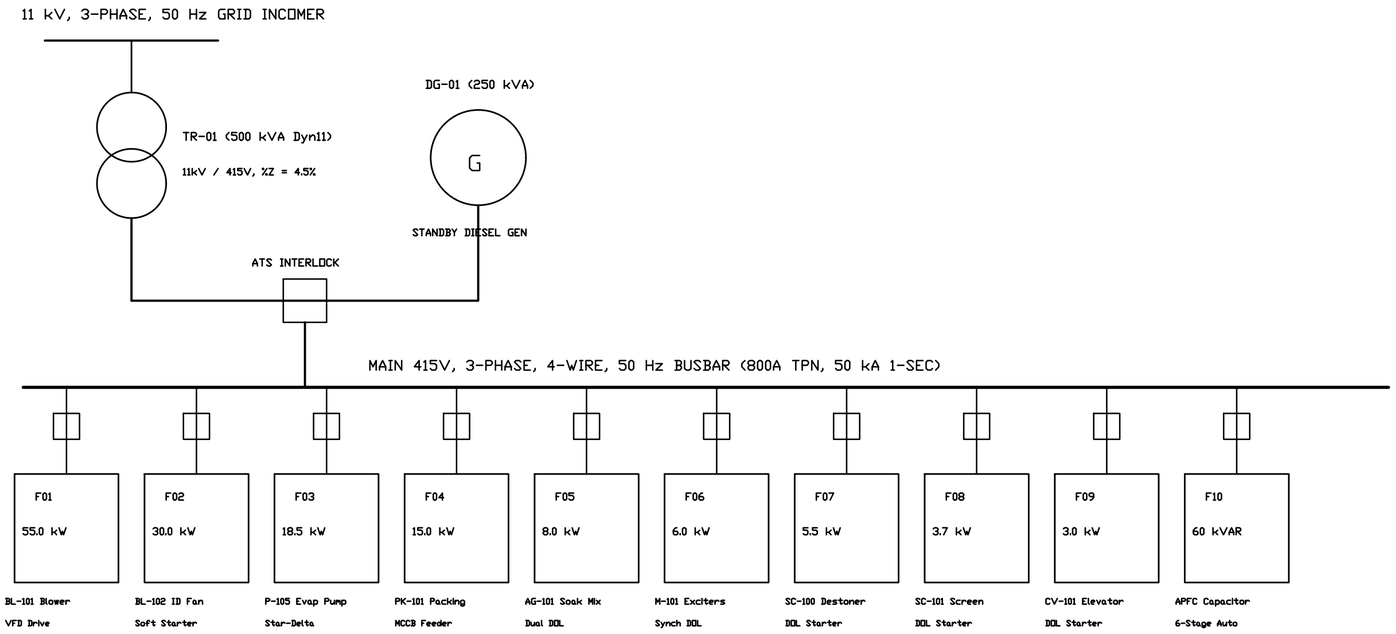
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

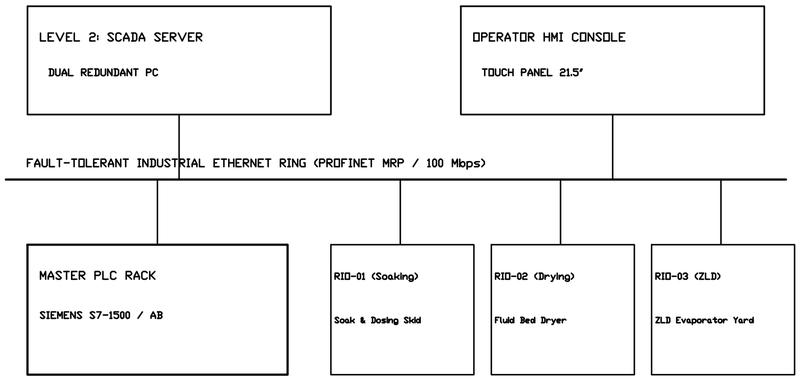
DWG-FIN-15T-010-3D: EV-101 TRIPLE-EFFECT ZLD EVAPORATOR 3D SOLID MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-010-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 20 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



INDUSTRIAL SCADA & PLC AUTOMATION ARCHITECTURE (ISA-95 / IEC 62443):



LEVEL 0 FIELD INTERFACE: HART 4-20mA Analog Loops, Modbus RS-485 Power Meters, VFDs on PROFINET

SAFETY INTEGRITY: Dual E-Stop Safety Relays (Cat 4 / SIL 3) hardwired to MCC Main Incomer Trips

ELECTRICAL LOAD SUMMARY, MOTOR FEEDER SIZING & SWITCHGEAR SCHEDULE:

1. TOTAL CONNECTED LOAD: 184.5 kW | DIVERSITY FACTOR: 0.70 | PEAK RUNNING DEMAND: 129.2 kW (131.8 kVA @ 0.98 PF).

2. POWER FACTOR CORRECTION: 60 kVAR Automatic APFC Bank maintains bus power factor at >= 0.98 lag (Penalty free).

3. TRANSFORMER SIZING: 500 kVA, 11kV/415V, Dyn11, ONAN cooling. Loading: 26.4% (Substantial headroom for Phase 2).

4. STANDBY GENERATION: 250 kVA (200 kW @ 0.8 PF) Diesel Generator with automatic mains failure (AMF) and ATS interlock.

5. UNINTERRUPTIBLE POWER SUPPLY (UPS): 10 kVA Online industrial true-sine UPS powering PLC, SCADA, and safety instruments.

6. SHORT CIRCUIT RATING: 50 kA for 1 second fault withstand rating on all MCC switchgear, busbars, and circuit breakers.

7. EARTHING & PROTECTION: Combined TN-S system with dedicated chemical copper earthing pits (<1.0 ohm resistance).

8. LIGHTNING PROTECTION: Early Streamer Emission (ESE) air terminal per IS/IEC 62305 with 45m protection radius.

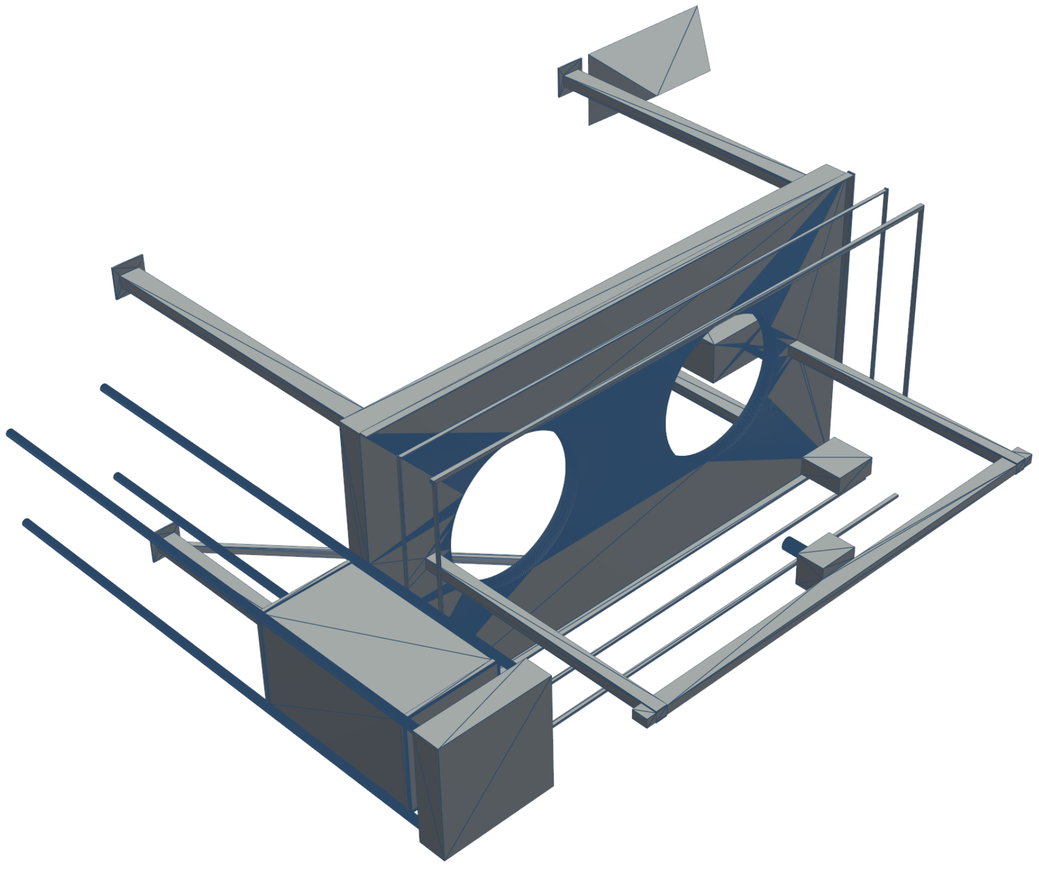
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

ELECTRICAL SINGLE-LINE DIAGRAM & AUTOMATION SCADA TOPOLOGY
DISCIPLINE: ELECTRICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-011	REV: V8.0	SHEET: 11 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PROJECT REF: FNQ-PR-2026-001 | DOC STATUS: **CONCEPTUAL FEED — PRELIMINARY**
GOVERNING STANDARDS: ASME SEC VIII DIV 1 / NFPA 68 / ISA-88 / FSSAI / IS 800



PRIMARY 3D SOLIDWORKS B-REP SOLID MODEL PROJECTION (SOLIDWORKS 2025 SP01.2 / VTK RENDER)

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Main Control Panel SCP-100: Form 4b CRCA enclosure (1,200 x 600 x 2,000mm) IP54 rated.
2. HMI: Siemens 21.5-inch Industrial Touch PC with WinCC Unified SCADA interface mounted at eye level.
3. Remote I/O: RIO-101 mounted on mezzanine platform; RIO-102 mounted on chemical dosing skid.
4. Communication network: Redundant Profinet Media Redundancy Protocol (MRP) industrial Ethernet ring.
5. Safety PLC: Siemens S7-1500F fail-safe controller managing emergency stops, interlocks, and burner trips.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

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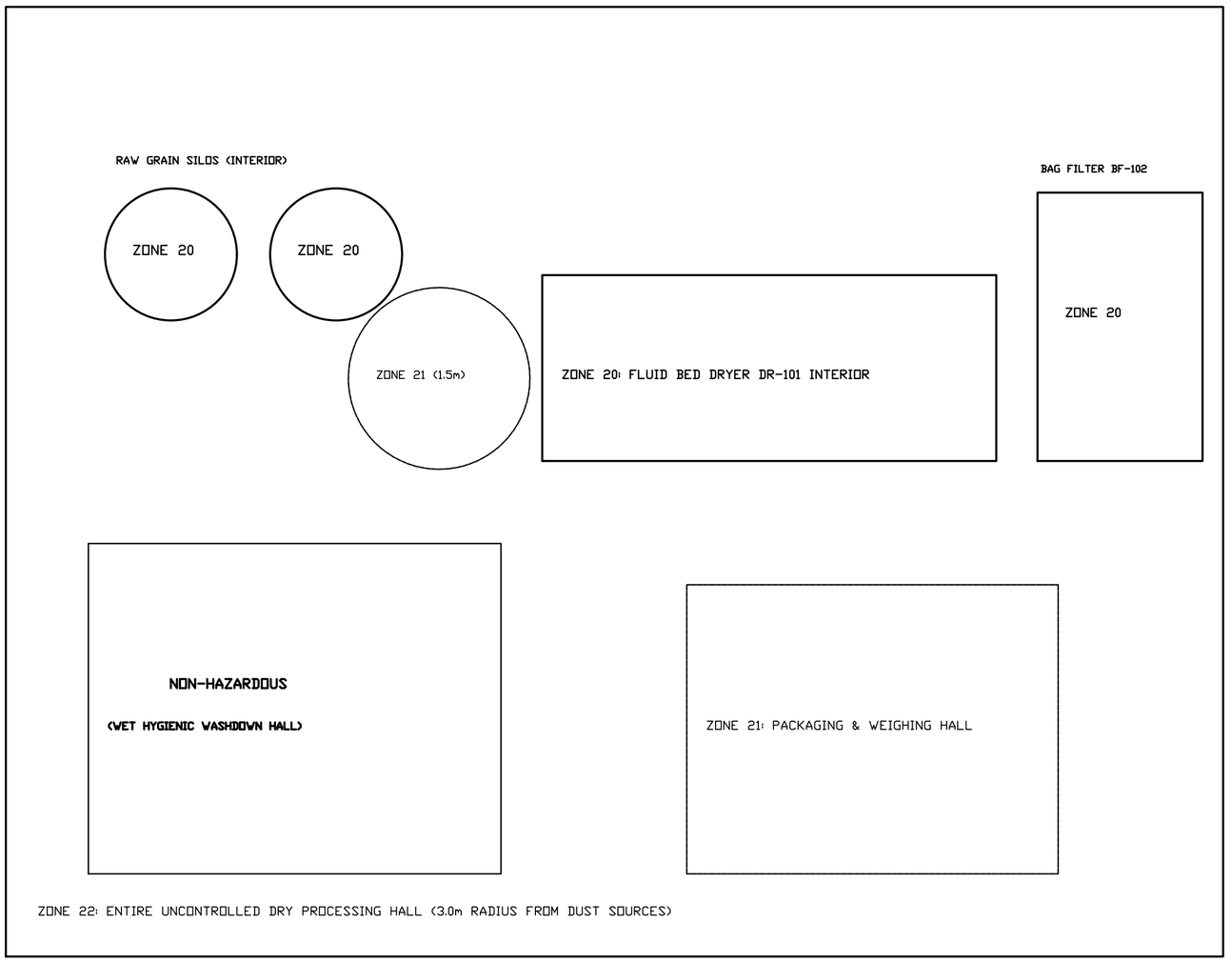
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-011-3D: MAIN SCADA & MOTOR CONTROL CENTER (MCC) PANEL 3D MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-011-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 22 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PROCESSING BUILDING - PLAN VIEW COMBUSTIBLE DUST ZONING (NFPA 652 / IEC 60079-10-2)



COMBUSTIBLE DUST CHARACTERISTICS (NFPA 652):

1. Material: Tanzanian Toor Dal Grain Dust (Cajanus cajan)

2. Dust Deflagration Class: St 1 (Kst: 85 bar-m/s, moderate)

3. Maximum Explosion Pressure Pmax: 7.6 bar g

4. Minimum Ignition Energy (MIE): > 30 mJ (No spark hazard)

5. Minimum Ignition Temp (MIT - Cloud): 430 deg C

6. Layer Ignition Temperature (LIT - 5mm): 260 deg C

7. Permissible Equipment Surface Temp: T120 deg C

   (Max 2/3 of MIT / 75K below LIT per IEC 60079-0)

8. Zone 20 Enclosures: IP65 Ex tb IIIC T120 deg C Da

9. Zone 21/22 Enclosures: IP55 Ex tc IIIC T120 deg C Dc

10. Explosion Relief: Dual certified NFPA 68 burst panels

    on DR-101 hood and flameless vents on BF-102.

11. Static Bonding: All pipes, ductwork, filter cages and

    vessels bonded to earthing grid (< 10 ohms resistance).

12. Housekeeping Standard: Scheduled vacuum cleaning

    preventing dust layer accumulation > 1.0 mm.

HAZARDOUS AREA ZONE CLASSIFICATION DEFINITIONS & APPLICABLE STANDARDS:

ZONE 20: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently (> 1,000 h/year). Equipment must be Category 1D / EPL Da certified.

ZONE 21: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally (10 to 1,000 h/year). Equipment must be Category 2D / EPL Db certified.

ZONE 22: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only (< 10 h/year). Category 3D / EPL Dc.

REGULATORY COMPLIANCE: NFPA 652 (Standard on the Fundamentals of Combustible Dust), NFPA 68 (Standard on Explosion Protection by Deflagration Venting), IEC 60079-10-2 (Explosive Atmospheres - Dust Area Classification), OSHA 1910.307.

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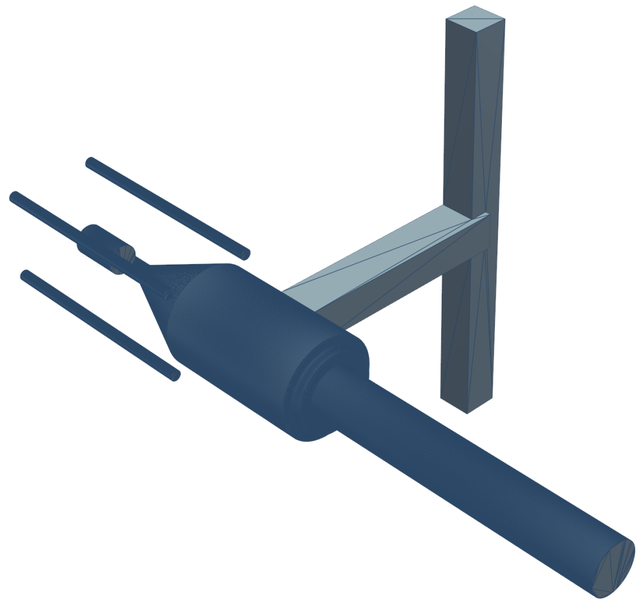
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

HAZARDOUS AREA CLASSIFICATION & DUST ZONING

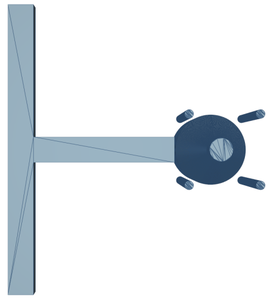
DISCIPLINE: SAFETY | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-012	REV: V8.0	SHEET: 12 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. SolidWorks B-Rep Part: CY-101_Cyclone_Separator.SLDPRT (29,722 vertices, 46,064 solid faces).
2. Cyclone dimensions: Barrel Dia 900mm, Cone Height 1,800mm, Vortex Finder Dia 450mm.
3. Explosion Protection: Top-mounted NFPA 68 rupture disc with weather cowl and burst detection sensor.
4. Rotary Airlock Valve: ATEX certified flame-proof rotary valve at cone discharge to prevent backfire.
5. Combustible Dust Classification: Zone 20 inside dryer/cyclone; Zone 21 within 1.5m; Zone 22 within 3.0m.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

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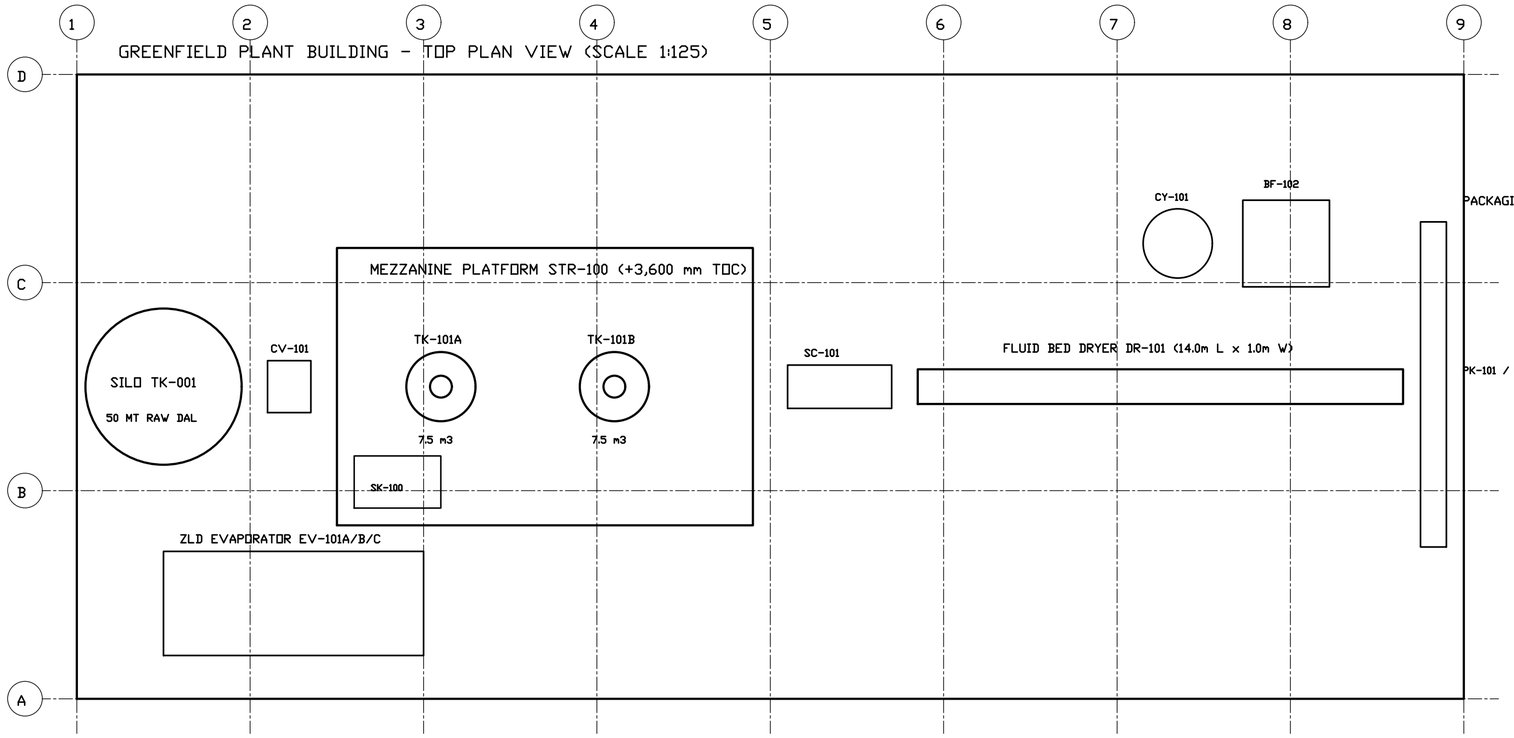
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

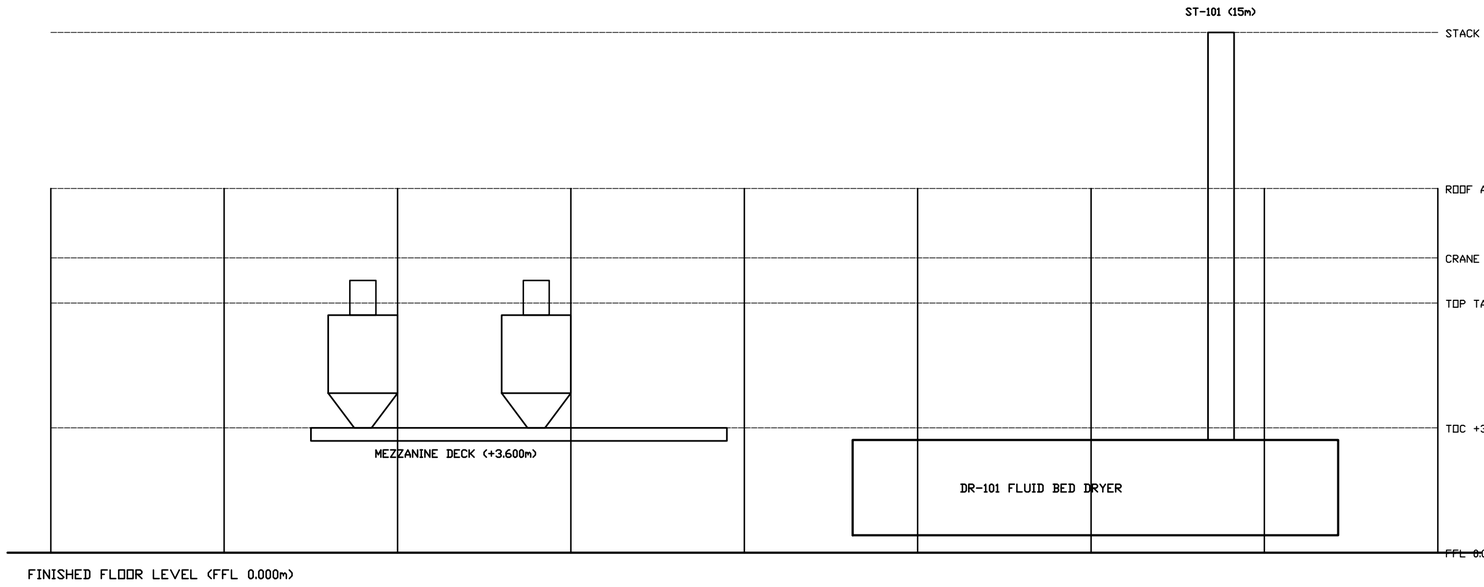
DWG-FIN-15T-012-3D: DUST COLLECTION CYCLONE & EXPLOSION VENTING 3D MODEL

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER: Sourish Senapati (FinnoaQ Agro-Industrial)
ACADEMIC GUIDANCE & RESEARCH: Dr. Archisman Ray (IIT Kharagpur)
CAD / SIMULATION PROVENANCE: SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14
FACILITY LOCATION & SCOPE: Greenfield Client Site | Gate G-A Experimental Hold

DWG NO: DWG-FIN-15T-012-3D
REV: 8.0
SHEET: PAGE 24 OF 45
SCALE: NTS / ISO A2





MASTER EQUIPMENT SCHEDULE & OPERATING WEIGHTS:

TK-001: Raw Grain Silo (50 MT)
Dim: Dia 4,500 x 6,500 | Elev: 0.000m
Weight: 58,500 kg | Power: 2.2 kW | MOC: Carbon Steel

CV-101: Bucket Elevator
Dim: 1,000 x 1,200 x 9,000 | Elev: 0.000m
Weight: 1,850 kg | Power: 3.7 kW | MOC: SS304 Contact

SC-100: Grain Cleaner & Destoner
Dim: 2,200 x 1,400 x 1,600 | Elev: 0.000m
Weight: 950 kg | Power: 2.2 kW | MOC: SS304 Casing

TK-101A: Soak Vessel A (7.5 m3)
Dim: Dia 2,000 x 3,450 | Elev: +3.600m
Weight: 9,850 kg | Power: 4.0 kW | MOC: AISI SS316L

TK-101B: Soak Vessel B (7.5 m3)
Dim: Dia 2,000 x 3,450 | Elev: +3.600m
Weight: 9,850 kg | Power: 4.0 kW | MOC: AISI SS316L

SK-100: Chemical Dosing Skid
Dim: 2,400 x 1,400 x 1,900 | Elev: +3.600m
Weight: 1,450 kg | Power: 1.5 kW | MOC: HDPE / SS316L

SC-101: Dewatering Screen
Dim: 3,000 x 1,200 x 1,200 | Elev: +3.600m
Weight: 1,250 kg | Power: 3.0 kW | MOC: AISI SS316L

DR-101: Fluid Bed Dryer
Dim: 14,000 x 1,000 x 2,800 | Elev: 0.000m
Weight: 6,850 kg | Power: 6.0 kW | MOC: AISI SS304

HE-101: Steam Air Heater
Dim: 1,800 x 1,200 x 1,500 | Elev: 0.000m
Weight: 1,200 kg | Power: N/A | MOC: Al-Finned SS

BL-101: Primary Air Blower
Dim: 1,600 x 1,400 x 1,400 | Elev: 0.000m
Weight: 680 kg | Power: 18.5 kW | MOC: Carbon Steel

CY-101: Cyclone Separators (2x)
Dim: Dia 1,200 x 4,000 | Elev: 0.000m
Weight: 1,700 kg | Power: N/A | MOC: AISI SS304

BF-102: Dust Collector Baghouse
Dim: 2,800 x 2,200 x 4,800 | Elev: 0.000m
Weight: 2,450 kg | Power: N/A | MOC: Carbon Steel

PK-101: Packaging Machine Line
Dim: 4,500 x 2,000 x 3,200 | Elev: 0.000m
Weight: 3,200 kg | Power: 7.5 kW | MOC: AISI SS304

EV-101: Triple Effect ZLD Evap
Dim: 7,500 x 3,000 x 4,500 | Elev: 0.000m
Weight: 5,500 kg | Power: 11.0 kW | MOC: AISI SS316L

TR-01: Substation Transformer
Dim: 2,200 x 1,600 x 2,000 | Elev: 0.000m
Weight: 2,800 kg | Power: 500 kVA | MOC: Oil Immersed

DG-01: Standby Generator Yard
Dim: 3,500 x 1,500 x 2,200 | Elev: 0.000m
Weight: 3,500 kg | Power: 200 kW | MOC: Acoustic Encl

DESIGN CODES & CIVIL STANDARDS:

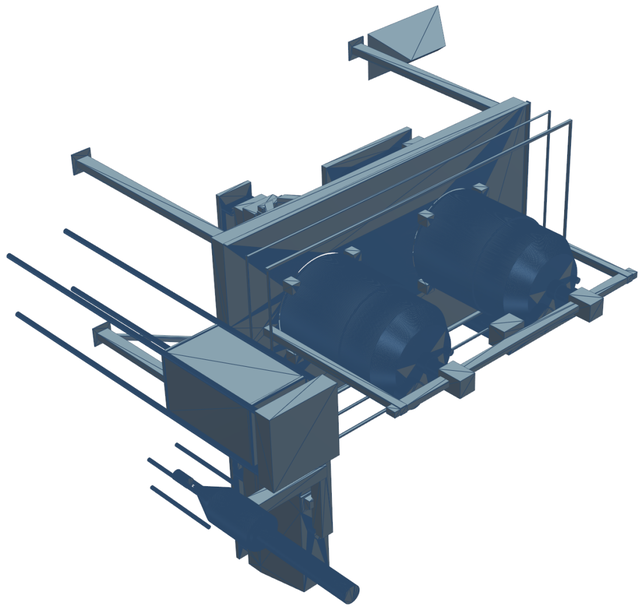
1. Civil Foundation: Safe bearing capacity 180 kN/m2 | IS 456 M25 grade concrete.
2. Structural Steel: IS 800:2007 | Columns ISMB 400 | Mezzanine Beams ISMB 300.
3. Design Loads: Process Mezzanine: 7.5 kN/m2 | Ground Floor: 25.0 kN/m2.
4. Total Operating Equipment Weight: 111.6 Tonnes (including grain & liquor fill).
5. Total Plant Structural Steel Weight: 27.08 Tonnes (Platforms, Ladders, Handrails).
6. Safety & Ergonomics: OSHA 1910.29 compliant walkways (1,000mm clear width).

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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

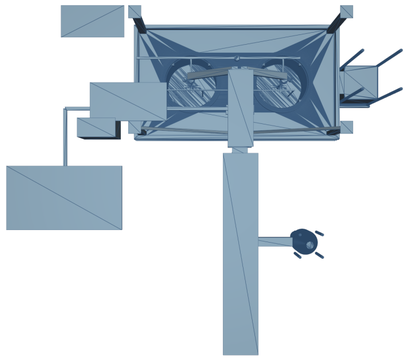
PLANT GENERAL ARRANGEMENT MASTER LAYOUT
DISCIPLINE: GENERAL ARRANGEMENT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-014	REV: V8.0	SHEET: 13 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

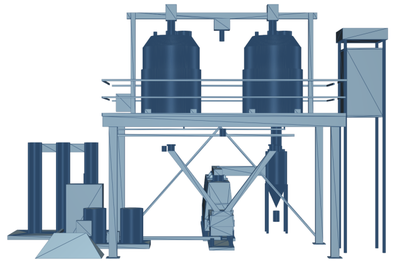
CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



VIEW A: PRIMARY 3D ISOMETRIC SOLID MODEL PROJECTION (METALLIC SHADING & EDGES)



VIEW B: ORTHOGRAPHIC FRONT ELEVATION



VIEW C: ORTHOGRAPHIC TOP PLAN VIEW

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. Master SolidWorks Assembly: PLANT-15T_General_Arrangement.SLDASM & STEP AP214 solid shell.
2. 12 Subsystems fully mated with zero gap: Soak Tanks, Agitators, Mezzanine, Chutes, Screen, Dryer, Cyclone, Dosing, Evaporator, Piping.
3. Total assembled solid mass: 27,078.89 kg (27.08 tonnes); Displaced volume: 27.0789 m3.
4. Complete balloon equipment callouts (1 to 10) matched to ASME/IS procurement BOM table.
5. Verified for structural rigidity, ergonomic operator clearances, and OSHA 1910 walkway compliance.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
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N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
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BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
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02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
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07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

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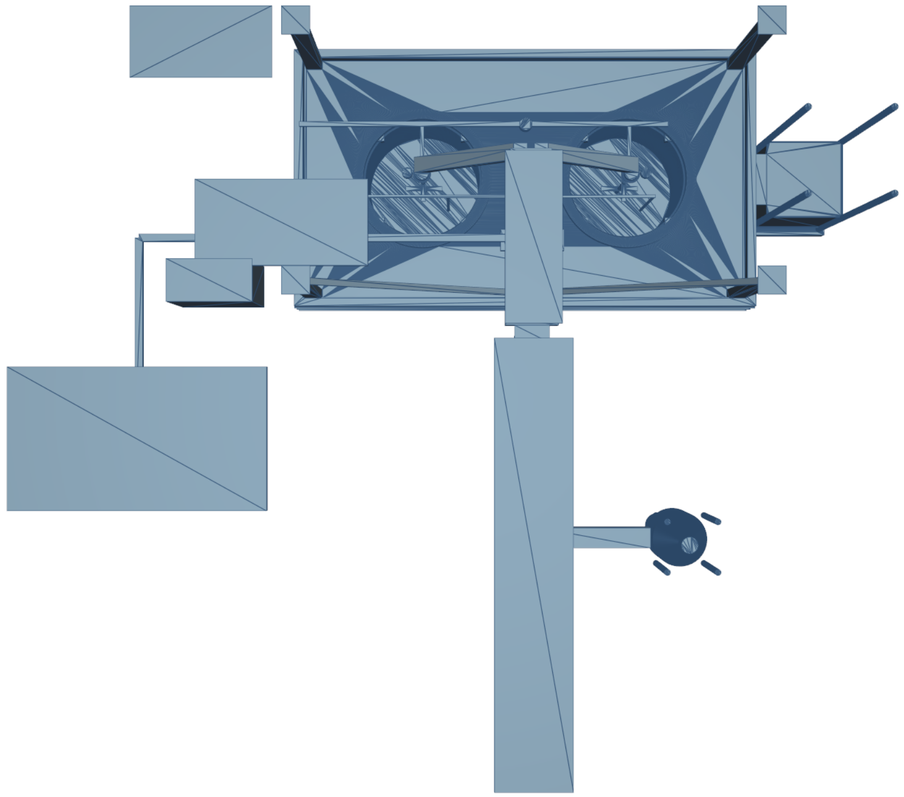
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-014-3D: 15 TPD PLANT MASTER ASSEMBLY 3D ISOMETRIC GENERAL ARRANGEMENT

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-014-3D**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	PAGE 26 OF 45
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PROJECT REF: FNQ-PR-2026-001 | DOC STATUS: **CONCEPTUAL FEED — PRELIMINARY**
GOVERNING STANDARDS: ASME SEC VIII DIV 1 / NFPA 68 / ISA-88 / FSSAI / IS 800



PRIMARY 3D SOLIDWORKS B-REP SOLID MODEL PROJECTION (SOLIDWORKS 2025 SP01.2 / VTK RENDER)

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. True orthographic front projection of PLANT-15T_General_Arrangement.SLDASM.
2. Critical elevations: Finished Floor Level = +0.000m; Mezzanine Deck = +4.200m; Agitator Drive Top = +6.850m.
3. Chute slope: 52.65 degrees gravity angle ensuring instantaneous dal slurry evacuation without bridging.
4. Dewatering screen deck elevation: Infeed lip at +1.450m; Discharge lip at +1.200m mating to dryer boot.
5. Fluid bed dryer air exhaust duct centerline: +3.850m rising to cyclone inlet at +4.650m.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

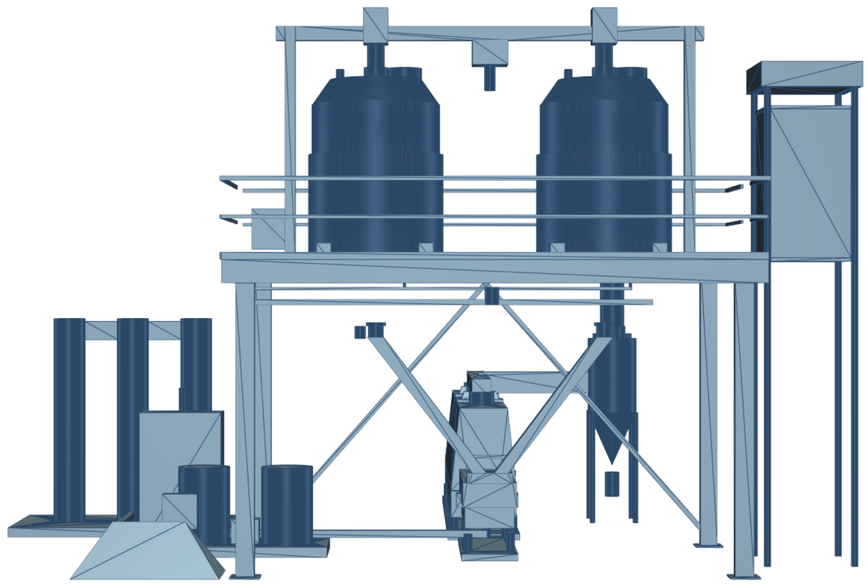
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

DWG-FIN-15T-014-B: PLANT MASTER ASSEMBLY ORTHOGRAPHIC FRONT ELEVATION

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-014-B**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	2 OF 2
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2



PRIMARY 3D SOLIDWORKS B-REP SOLID MODEL PROJECTION (SOLIDWORKS 2025 SP01.2 / VTK RENDER)

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. True orthographic top plan projection of PLANT-15T_General_Arrangement.SLDASM.
2. Primary building bay grid: 6.0m column centers along X-axis; 7.0m column centers along Z-axis.
3. Maintenance clearances: Minimum 1,200mm clear walkway around all rotary equipment and tanks.
4. Central aisle: 3,000mm forklift access corridor for material handling and maintenance access.
5. Mezzanine deck cutouts: Dual circular penetrations Dia 1,850mm with under-deck collar frames.

EQUIPMENT NOZZLE SCHEDULE & PROCESS CONNECTIONS

Tag	Size	Rating	Facing	Service	MOC
N1	DN150 6"	150#	RF	Grain Infeed / Manway	SS316L
N2	DN80 3"	150#	RF	Hydration Liquor Feed	SS316L
N3	DN50 2"	150#	RF	CIP Spray Ball Header	SS316L
N4	DN40 1.5"	150#	RF	Chemical Dosing Trunk	SS316L
N5	DN150 6"	150#	RF	Bottom Slurry Discharge	SS316L
N6	DN50 2"	150#	RF	Dimple Jacket Steam In	SS304
N7	DN50 2"	150#	RF	Jacket Condensate Drain	SS304
N8	DN80 3"	150#	RF	Vessel Overflow & Vent	SS316L

BILL OF MATERIALS & MECHANICAL COMPONENT SCHEDULE

Item	Component Description	Qty	Material	Spec / Rating
01	Cylindrical Vessel Shell (Dia 1,800mm, H=2,200mm)	1	SS316L	6mm Wall, ASME Sec VIII Div 1
02	60-Degree Conical Bottom Transition Cone	1	SS316L	8mm Wall, Knuckle Radius 150mm
03	Dimple Steam Jacket (OD 1,900mm x 1,400mm H)	1	SS304	10 bar Design Pres, 185 deg C
04	Heavy Structural Vessel Mounting Lugs (4 No.)	4	IS 2062	25mm Thick with SS304 cladding
05	Sanitary Hydrofoil Impeller (3-Blade, Dia 700mm)	1	SS316L	Electropolished Ra <= 0.4 um
06	Gearmotor Drive Unit (SEW-Eurodrive Helical Bevel)	1	Cast Iron	5.5 kW, 415V, IP66, 48 RPM Output
07	Double Mechanical Seal (Food Grade, Water Flushed)	1	SiC / Carbon	FDA / USP Class VI EPDM elastomers
08	Bottom Slurry Automated Butterfly Valve (FCV-101)	1	SS316L	DN150, Pneumatic Double-Acting

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

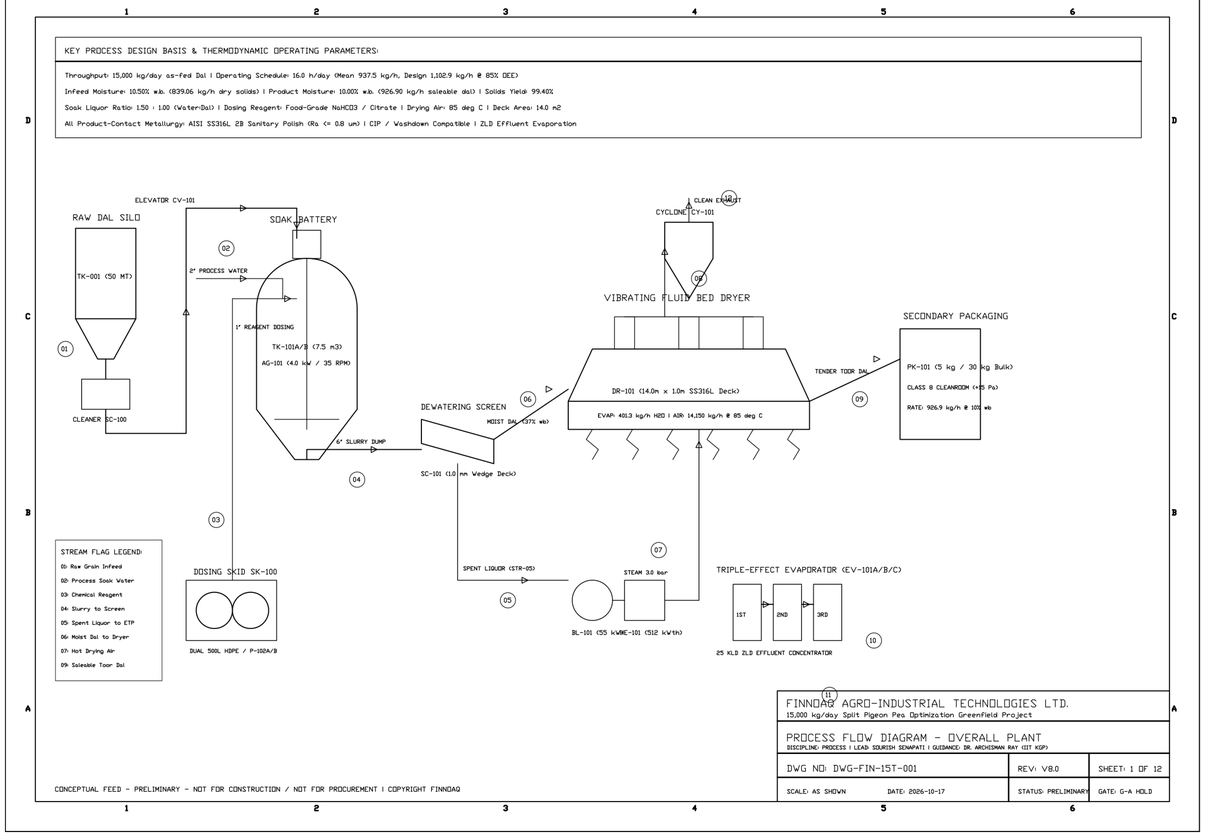
DWG-FIN-15T-014-C: PLANT MASTER ASSEMBLY ORTHOGRAPHIC TOP PLAN VIEW

SOLIDWORKS 2025 SP01.2 PARAMETRIC B-REP SOLID MODEL VERIFICATION

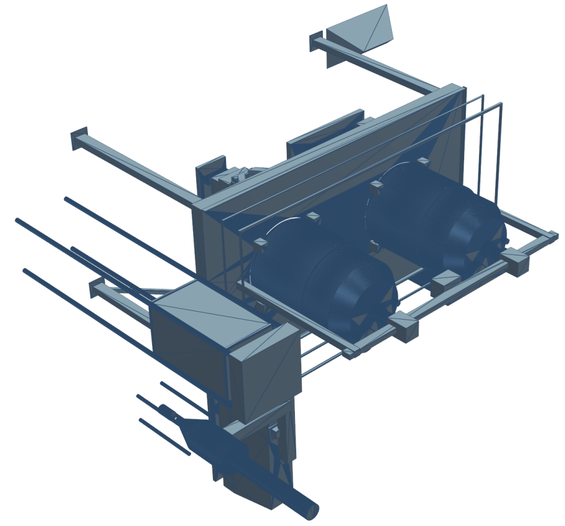
LEAD CHEMICAL & PROCESS ENGINEER: Sourish Senapati (FinnoaQ Agro-Industrial)
ACADEMIC GUIDANCE & RESEARCH: Dr. Archisman Ray (IIT Kharagpur)
CAD / SIMULATION PROVENANCE: SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14
FACILITY LOCATION & SCOPE: Greenfield Client Site | Gate G-A Experimental Hold

DWG NO: DWG-FIN-15T-014-C
REV: 8.0
SHEET: 2 OF 2
SCALE: NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

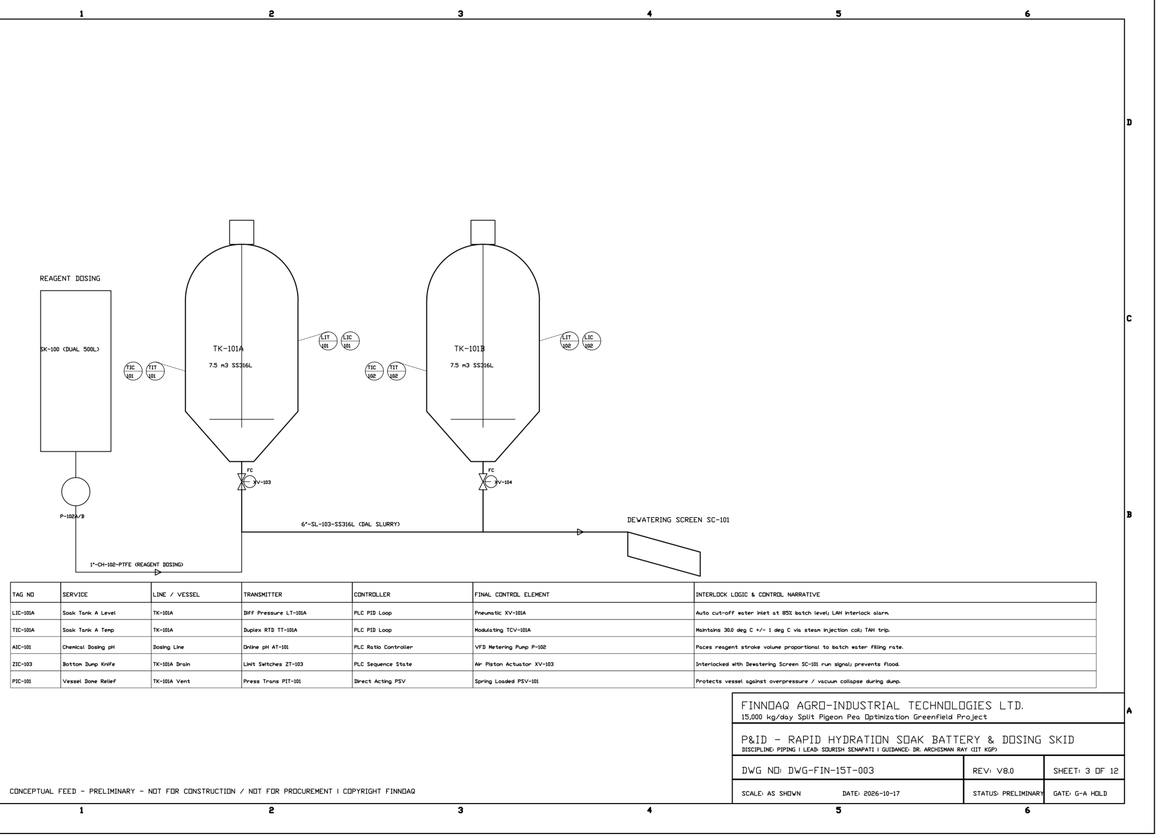
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: PROCESS FLOW DIAGRAM & 3D PLANT GENERAL ARRANGEMENT

INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-001-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



TAG NO	SERVICE	LINE / VESSEL	TRANSMITTER	CONTROLLER	FINAL CONTROL ELEMENT	INTERLOCK LOGIC & CONTROL NARRATIVE
LIC-101A	Soak Tank A Level	TK-101A	Diff Pressure LT-101A	PLC PID Loop	Pneumatic XV-101A	Auto cut-off water inlet at 85% batch level; LAH interlock alarm.
TIC-102A	Soak Tank A Temp	TK-101A	Duplex RTD TT-102A	PLC PID Loop	Modulating TCV-101A	Maintains 368 deg C +/- 1 deg C via steam injection coil; TAH trip.
AIC-101	Chemical Dosing pH	Dosing Line	Online pH AT-101	PLC Ratio Controller	VFD Metering Pump P-102	Paces reagent stroke volume proportional to batch water filling rate.
ZIC-103	Bottom Dump Knife	TK-101A Drain	Limit Switches ZT-103	PLC Sequence State	Air Piston Actuator XV-103	Interlocked with Dewatering Screen SC-101 run signal; prevents flood.
PIC-101	Vessel Dome Relief	TK-101A Vent	Press Trans PIT-101	Direct Acting PSV	Spring Loaded PSV-101	Protects vessel against overpressure / vacuum collapse during dump.

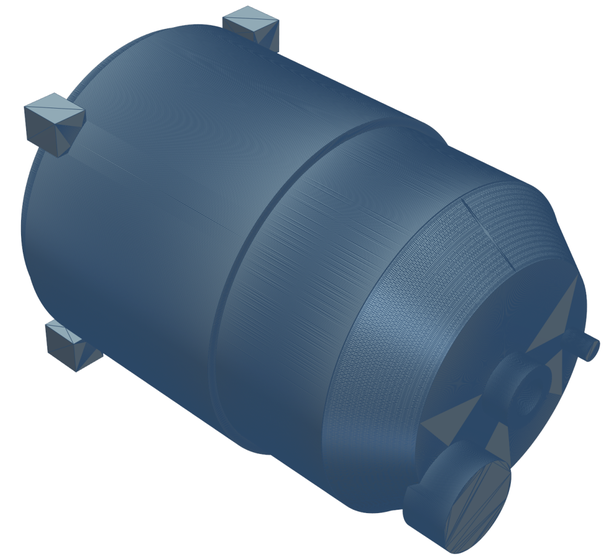
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - RAPID HYDRATION SOAK BATTERY & DOSING SKID
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-003 | REV: V8.0 | SHEET: 3 OF 12
SCALE: AS SHOWN | DATE: 2026-10-17 | STATUS: PRELIMINARY | GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

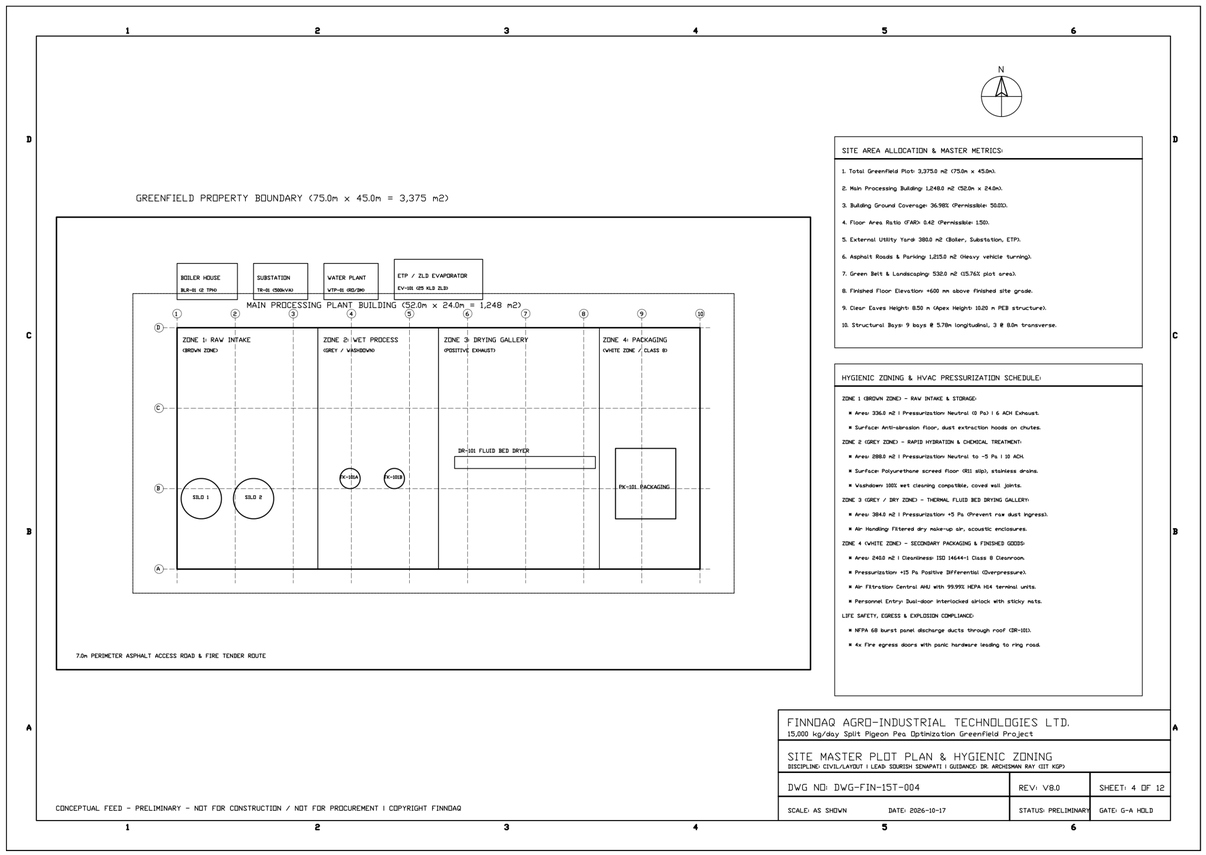
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: SOAKING SECTION P&ID & TK-101 SOAK TANK 3D SOLID MODEL

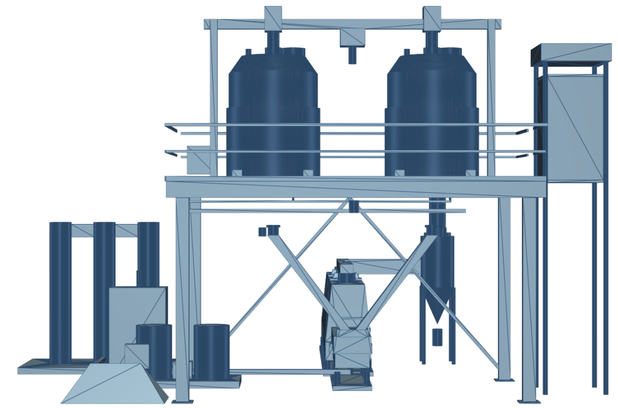
INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-003-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

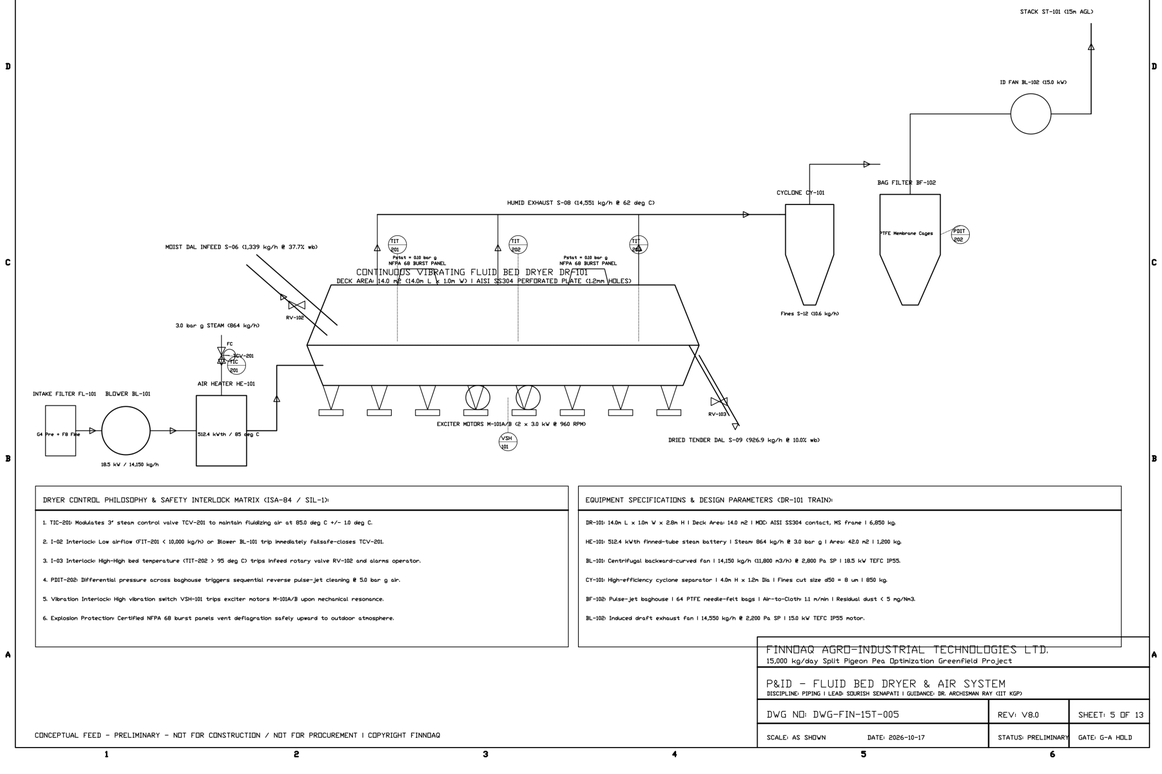
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: SITE PLOT PLAN & 3D MASTER BUILDING LAYOUT

INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-004-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



DRYER CONTROL PHILOSOPHY & SAFETY INTERLOCK MATRIX (ISA-84 / SIL-1)

1. TIC-201: Modulates 3" steam control valve TCV-201 to maintain fluidizing air at 85.0 deg C +/- 1.0 deg C.
2. I-01 Interlock: Low airflow (FIT-201 < 10,000 kg/h) or Blower BL-101 trip immediately fail-safe-closes TCV-201.
3. I-03 Interlock: High-High bed temperature (TIT-202 > 95 deg C) trips infeed rotary valve RV-102 and alarms operator.
4. PDIT-202: Differential pressure across baghouse triggers sequential reverse pulse-jet cleaning @ 5.0 bar g air.
5. Vibration Interlock: High vibration switch VSH-101 trips exciter motors M-101A/B upon mechanical resonance.
6. Explosion Protection: Certified NFPA 68 burst panels vent deflagration safely upward to outdoor atmosphere.

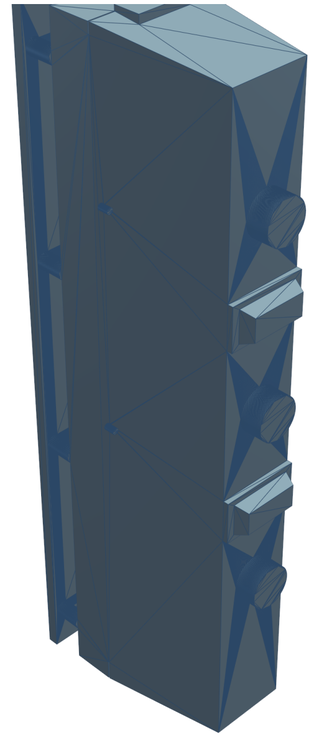
EQUIPMENT SPECIFICATIONS & DESIGN PARAMETERS (DR-101 TRAIN)

DR-101: 14.0m L x 1.0m W x 2.8m H | Deck Area: 14.0 m2 | MOC: AISI SS304 contact, MS frame | 6,850 kg.
HE-101: 512.4 kWth finned-tube steam battery | Steam: 864 kg/h @ 3.0 bar g | Area: 42.0 m2 | 1,200 kg.
BL-101: Centrifugal backward-curved fan | 14,150 kg/h (11,800 m3/h) @ 2,800 Pa SP | 18.5 kW TEFC IP55.
CY-101: High-efficiency cyclone separator | 4.0m H x 1.2m Dia | Fines cut size d50 = 8 um | 850 kg.
BF-102: Pulse-jet baghouse | 64 PTFE needle-felt bags | Air-to-Cloth: 1.1 m/min | Residual dust < 5 mg/Nm3.
BL-102: Induced draft exhaust fan | 14,550 kg/h @ 2,200 Pa SP | 150 kW TEFC IP55 motor.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.			
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project			
P&ID - FLUID BED DRYER & AIR SYSTEM			
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)			
DWG NO: DWG-FIN-15T-005	REV: V8.0	SHEET: 5 OF 13	
SCALE: AS SHOWN	DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

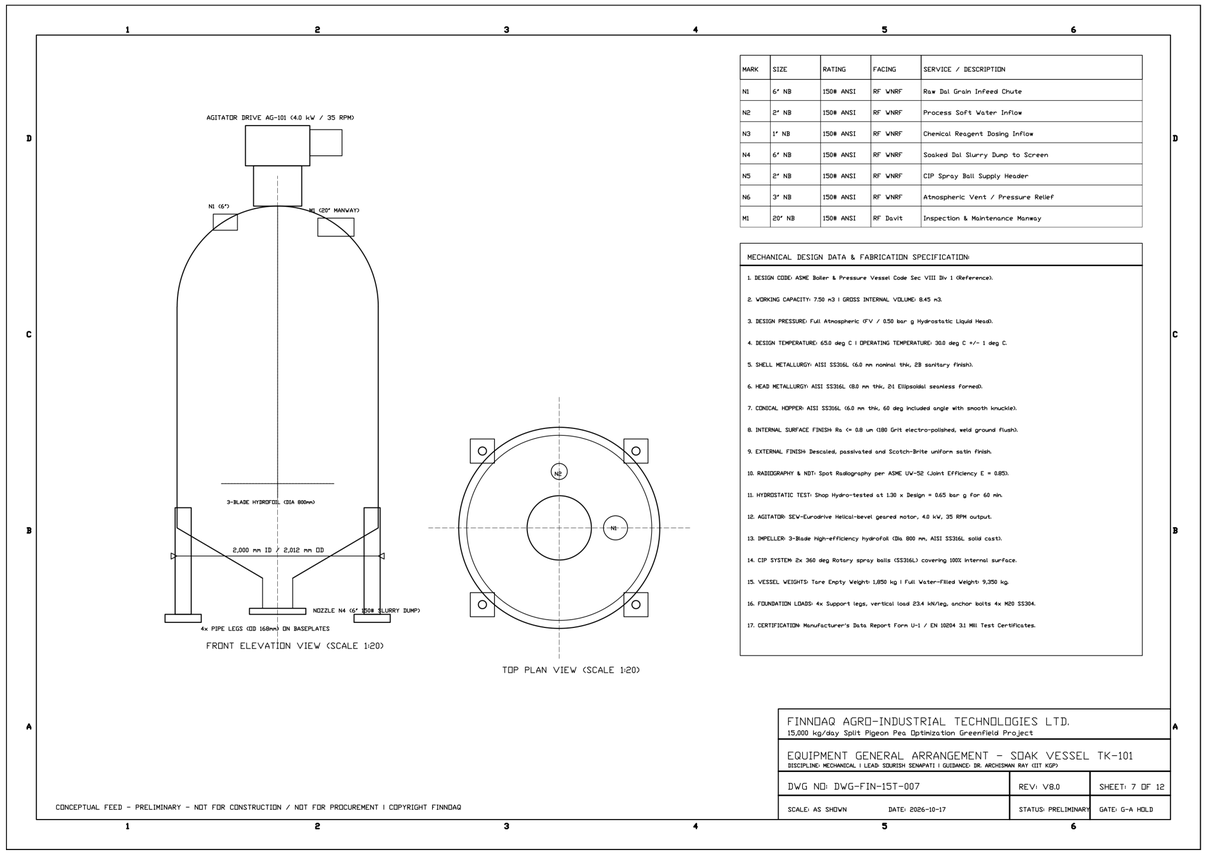
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: DRYER SECTION P&ID & DR-101 FLUID BED DRYER 3D MODEL

INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-005-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



MARK	SIZE	RATING	FACING	SERVICE / DESCRIPTION
N1	6" NB	150# ANSI	RF WNRF	Raw Dal Grain Infeed Chute
N2	2" NB	150# ANSI	RF WNRF	Process Soft Water Inflow
N3	1" NB	150# ANSI	RF WNRF	Chemical Reagent Dosing Inflow
N4	6" NB	150# ANSI	RF WNRF	Soaked Dal Slurry Dump to Screen
N5	2" NB	150# ANSI	RF WNRF	CIP Spray Ball Supply Header
N6	3" NB	150# ANSI	RF WNRF	Atmospheric Vent / Pressure Relief
M1	20" NB	150# ANSI	RF Davit	Inspection & Maintenance Manway

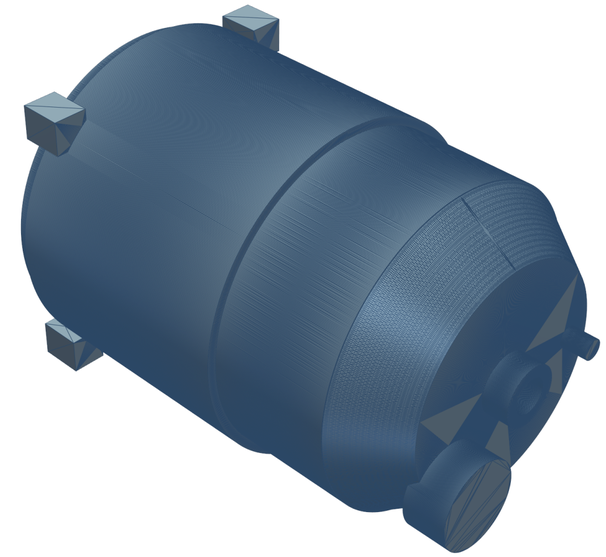
MECHANICAL DESIGN DATA & FABRICATION SPECIFICATION:

1. DESIGN CODE: ASME Boiler & Pressure Vessel Code Sec VIII Div 1 (Reference).
2. WORKING CAPACITY: 7.50 m3 | GROSS INTERNAL VOLUME: 8.45 m3.
3. DESIGN PRESSURE: Full Atmospheric (FV / 0.50 bar g Hydrostatic Liquid Head).
4. DESIGN TEMPERATURE: 65.0 deg C | OPERATING TEMPERATURE: 30.0 deg C +/- 1 deg C.
5. SHELL METALLURGY: AISI SS316L (6.0 mm nominal thk, 2B sanitary finish).
6. HEAD METALLURGY: AISI SS316L (8.0 mm thk, 2:1 Ellipsoidal seamless formed).
7. CONICAL HOPPER: AISI SS316L (6.0 mm thk, 60 deg included angle with smooth knuckle).
8. INTERNAL SURFACE FINISH: Ra <= 0.8 um (180 Grit electro-polished, weld ground flush).
9. EXTERNAL FINISH: Descaled, passivated and Scotch-Brite uniform satin finish.
10. RADIOGRAPHY & NDT: Spot Radiography per ASME UW-52 (Joint Efficiency E = 0.85).
11. HYDROSTATIC TEST: Shop Hydro-tested at 1.30 x Design = 0.65 bar g for 60 min.
12. AGITATOR: SEW-Eurodrive Helical-bevel geared motor, 4.0 kW, 35 RPM output.
13. IMPELLER: 3-Blade high-efficiency hydrofoil (Dia 800 mm, AISI SS316L, solid cast).
14. CIP SYSTEM: 2x 360 deg Rotary spray balls (SS316L) covering 100% internal surface.
15. VESSEL WEIGHTS: Tare Empty Weight: 1,850 kg | Full Water-Filled Weight: 9,350 kg.
16. FOUNDATION LOADS: 4x Support legs, vertical load 23.4 kN/leg, anchor bolts 4x M20 SS304.
17. CERTIFICATION: Manufacturer's Data Report Form U-1 / EN 10204 3.1 Mill Test Certificates.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD. 15,000 kg/day Split Pigeon Pea Optimization Greenfield Project		
EQUIPMENT GENERAL ARRANGEMENT – SOAK VESSEL TK-101 DISCIPLINE: MECHANICAL	LEAD: SOURISH SENAPATI	GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)
DWG NO: DWG-FIN-1ST-007	REV: V8.0	SHEET: 7 OF 12
SCALE: AS SHOWN	DATE: 2026-10-17	STATUS: PRELIMINARY / GATE: G-A HOLD

CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

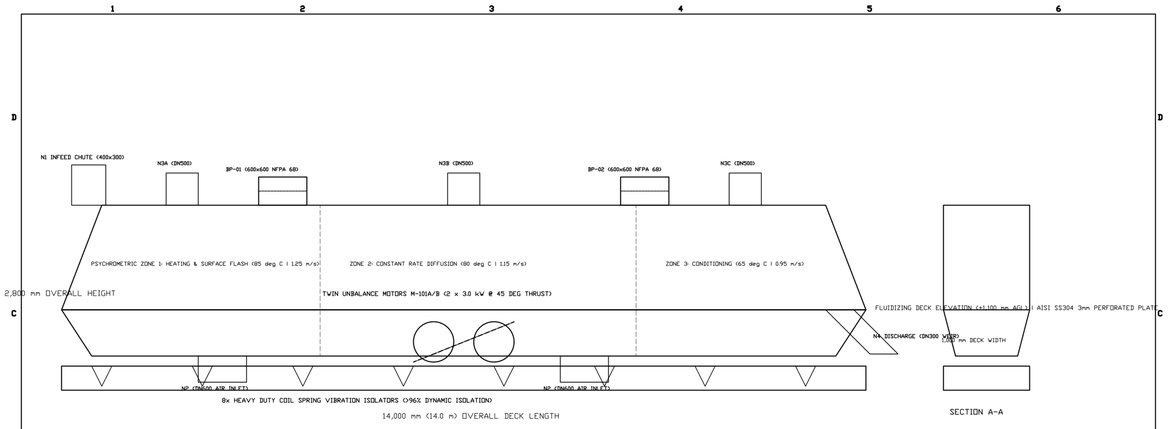
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: TK-101 SOAK TANK MECHANICAL GA & 3D B-REP SOLID MODEL

INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-007-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



MARK	SERVICE	SIZE	RATING	FACING	LOCATION
N1	Moist Dal Infeed	400x300	FLANGED	FLAT FACE	Top Hood Infeed End
N2A/B	Hot Air Supply	2x DN600	150#	RAISED FACE	Bottom Plenum Underside
N3A-C	Exhaust Air Outlets	3x DN500	150#	FLAT FACE	Top Hood Crown
N4	Dry Dal Discharge	DN300	150#	RAISED FACE	Discharge End Retention Weir
N5	Plenum Fines Drain	DN100	150#	RAISED FACE	Plenum Invert Low Point
M1-M4	Inspection Doors	4x 500x400	HINGED	FOOD GASKET	Drive & Non-Drive Sides
BP-01/02	NFPA 68 Burst Panels	2x 600x600	SS COMPOSITE	ROOF DUCTED	Expansion Hood Crown

MECHANICAL, VIBRATION DYNAMICS & NFPA 68 SAFETY SPECIFICATION:

1. Equipment Tag: DR-101 | Service: Continuous Vibrating Fluid Bed Dal Conditioning & Drying.
2. Effective Deck Dimensions: 14,000 mm Length x 1,000 mm Width (Effective Area: 14.0 m2).
3. Material of Construction: All product contact parts SS304 0Ra < 0.8 um sanitary polish). Non-contact IS2062 CS.
4. Fluidizing Air Rate: 14,150 kg/h (11,800 m3/h @ 85 deg C) | Superficial Velocity: 1.15 m/s (Umf = 0.85 m/s).
5. Water Evaporation Duty: 412.34 kg/h H2O | Thermal Heating Duty: 512.4 kWth via Steam Coil HE-101.
6. Vibration Dynamics: Stroke amplitude 3.5 mm @ 16 Hz (960 RPM); Twin unbalance motors 2 x 3.0 kW @ 45 deg.
7. Isolation Efficiency: 8x Heavy-duty helical coil springs with rubber snubbers (>96% dynamic isolation).
8. Residence Time: 18.0 - 22.0 minutes plug-flow with adjustable discharge overflow retention weir.
9. NFPA 68 EXPLOSION RELIEF: Dual 600x600 mm composite rupture panels (Pred = 0.2 bar g, Pstat = 0.1 bar g).
   - Sized for St-1 Agricultural Pulse Dust (Kst = 85 bar-m/s, Pmax = 7.8 bar g) with ducted vent to building exterior.
10. Safety Hold: Final dynamic balancing, spring resonance validation and FAT testing required prior to site release.

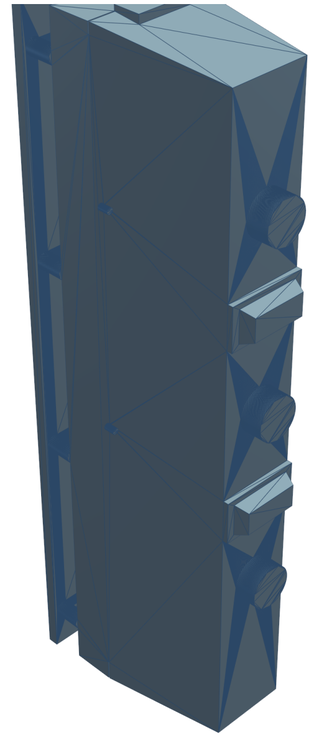
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - FLUID BED DRYER DR-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-008	REV: V8.0	SHEET: 8 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

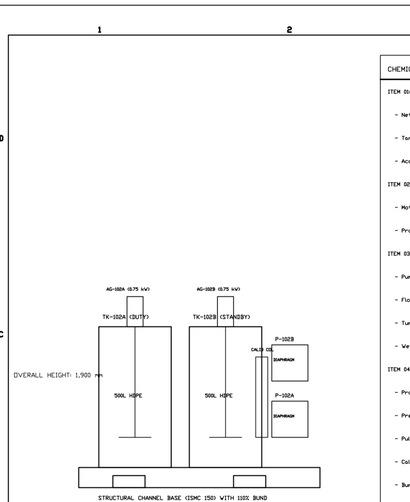
Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: DR-101 DRYER MECHANICAL GA & 3D B-REP SOLID MODEL
INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-008-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



CHEMICAL REAGENT DOSING SKID SPECIFICATION & BILL OF MATERIALS (SK-100):

ITEM 01: Solution Preparation Tanks TK-102A/B (2 Units, Duty / Standby)

- Net Working Volume: 500 Liters each | Material: UV-Stabilized Food-Grade Cross-Linked HDPE (FDA 21 CFR 177.1520).
- Tank Geometry: 800 mm Dia x 1,100 mm Straight Shell Height | Conical bottom with 1.5" drain valve.
- Accessories: Hinged powder loading hatch with safety interlock, dust screen, ultrasonic level transmitter LT-102.

ITEM 02: High-Shear Agitator Mixers AG-102A/B (2 Units)

- Motor: 0.75 kW, 415V, 3-Phase, 50 Hz, IP55, TEFC | Shaft & Impeller: AISI SS316L 4-Blade Marine Propeller.
- Process Service: Rapid dissolution of food-grade Sodium Bicarbonate / Trisodium Citrate salts @ 25-35 deg C.

ITEM 03: Precision Metering Chemical Pumps P-102A/B (2 Units, Duty / Standby)

- Pump Type: Positive Displacement Mechanically Actuated PTFE Diaphragm Dosing Pumps.
- Flow Capacity: 5.0 to 50.0 L/h (Normal Operating Point: 14.06 L/h for 1.0 wt% reagent liquor recipe).
- Turndown: 10:1 Turndown ratio with micrometer stroke length adjuster + 4-20mA VFD speed modulation.
- Wetted Metallurgy: PVDF Liquid End, Ceramic Double Ball Check Valves, Pure Virgin PTFE Diaphragm.

ITEM 04: Skid Piping, Instrumentation & Safety Systems

- Process Piping: 1" NB Schedule 10S AISI SS316L piping with sanitary tri-clamp fittings.
- Pressure Relief Valves: PSV-102A/B Set @ 4.0 bar g with full recirculation bypass to suction tank.
- Pulsation Dampener: Bladder-type 0.5L PVDF dampener smoothing discharge pulsation to < +/-2.5%.
- Calibration Column: 1,000 mL Borosilicate glass cylinder with 3-way isolation valve for flow rate verification.
- Bund Containment: 110% Spill containment capacity (600L) with optical liquid leakage switch LS-102 interlocked to ESD.

SKID NOZZLE SCHEDULE & UTILITY INTERFACE SCHEDULE:

NOZZLE	SERVICE	SIZE	RATING	FACING	METALLURGY	CONNECTED TO
N1	Soft Water Supply Infeed	1.5" NB	ANSI 150#	Flat Face	AISI SS316L	WTP-01 Header @ 2.5 bar g
N2	Chemical Dosing Discharge	1.0" NB	ANSI 150#	Raised Face	AISI SS316L	Soak Battery TK-101 Feed Spool
N3	Containment Bund Drain	2.0" NB	ANSI 150#	Flat Face	AISI SS304	ETP Equalization Pit TK-104
E1	Main Electrical Power Feed	4-Core	415V / 50Hz	Glanded	Armoured Cu	MCC-01 Feeder Feed-04 (3.5 kW)
I1	Automation Signal Trunk	12-Pair	4-20mA / DI	Shielded	Tinned Cu	Remote I/O Panel RIO-02 (Modbus TCP)

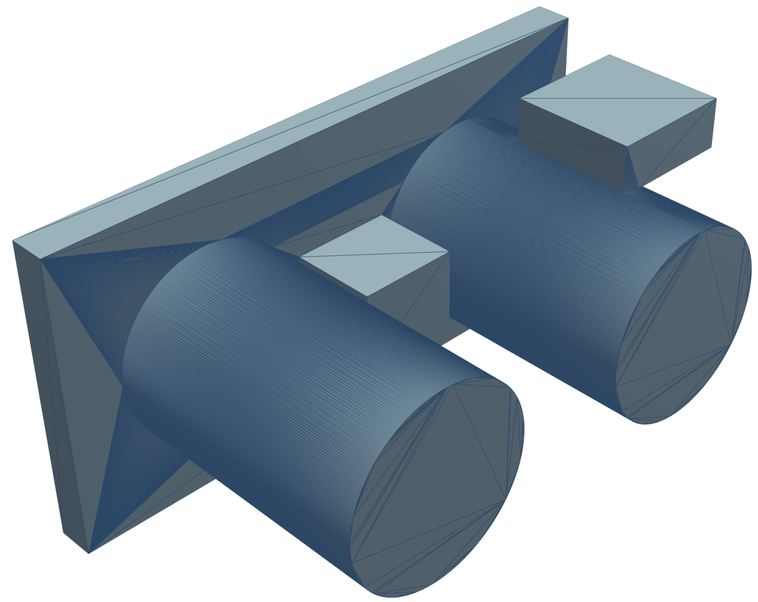
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GA - CHEMICAL DOSING SKID SK-100
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-009	REV: V8.0	SHEET: 9 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

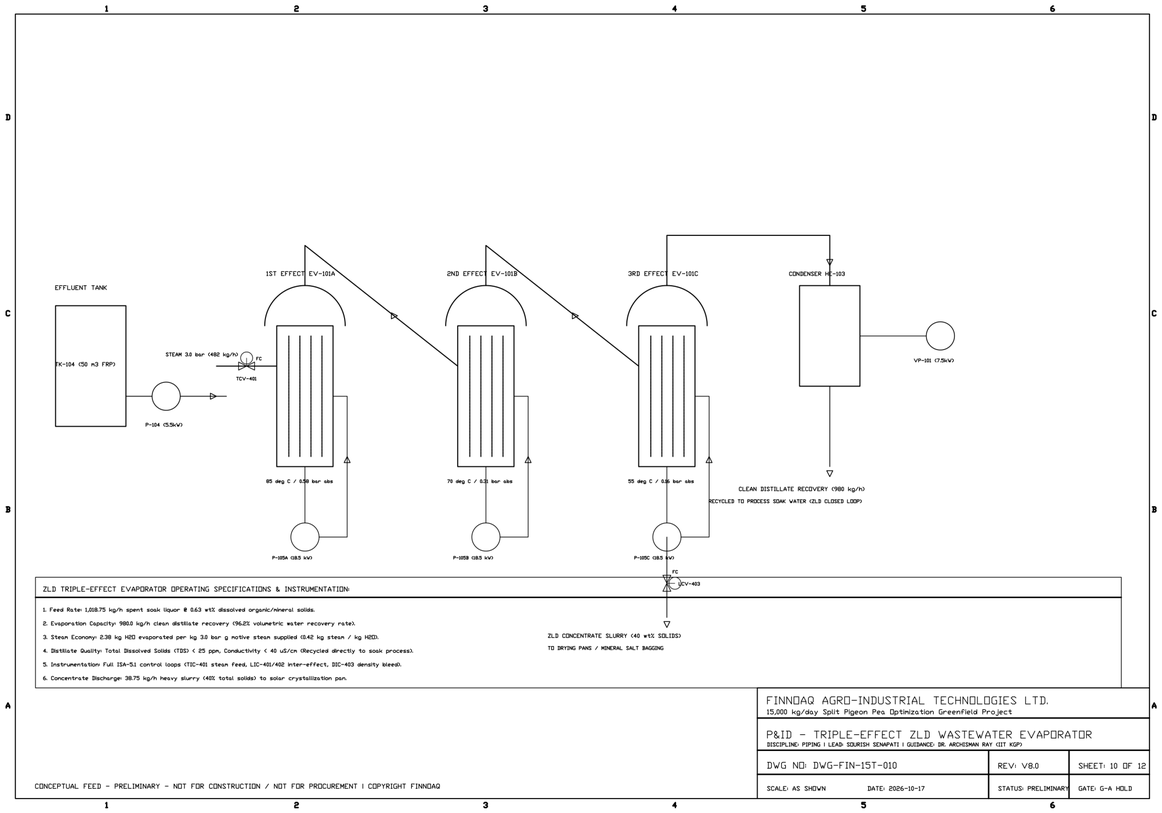
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: SK-100 CHEMICAL DOSING SKID GA & 3D SOLID MODEL

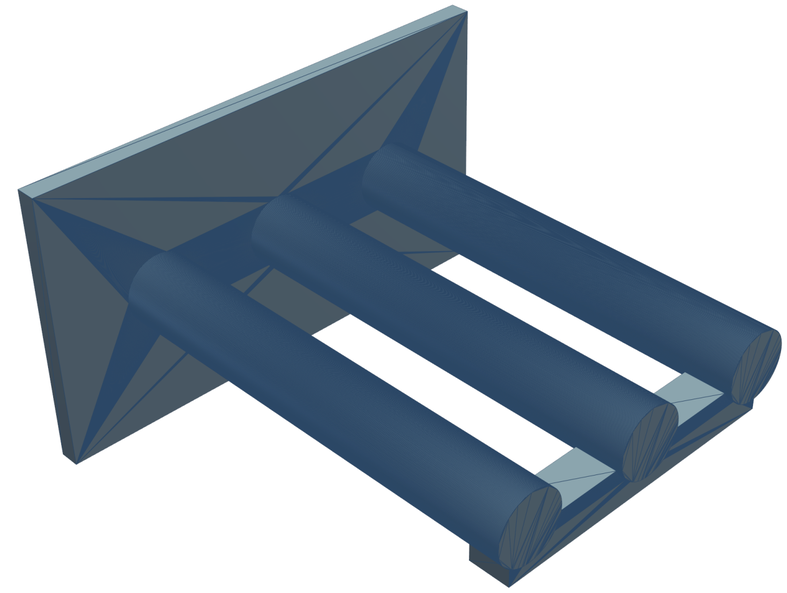
INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-009-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

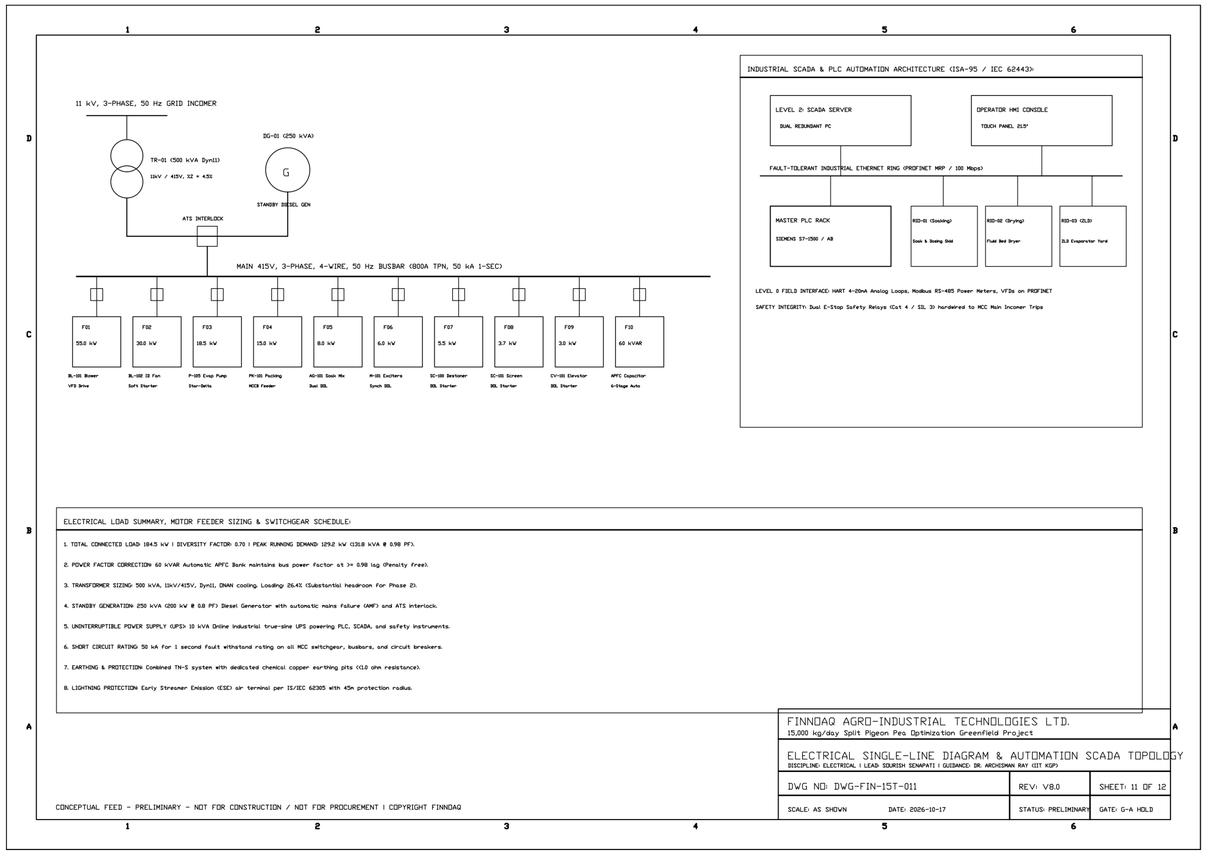
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: EV-101 ZLD EVAPORATOR P&ID & 3D SOLID MODEL

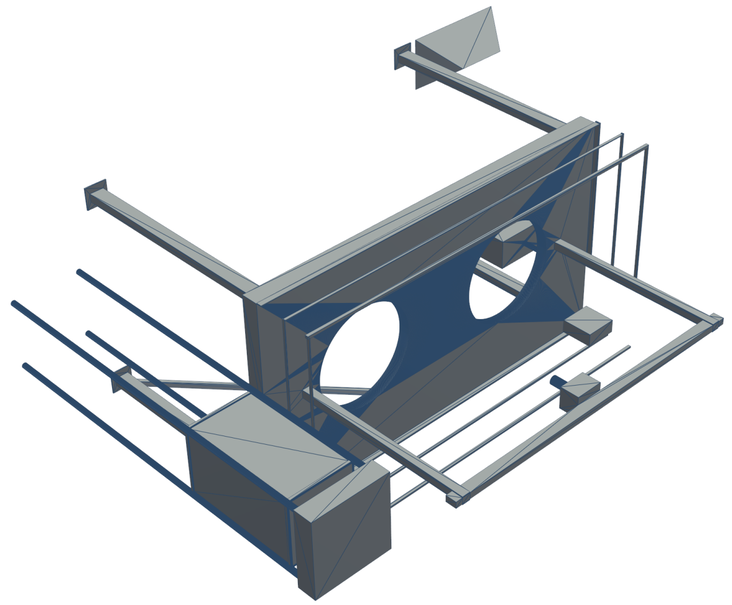
INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-010-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
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Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: ELECTRICAL SLD & MAIN SCADA / MCC ENCLOSURE 3D MODEL

INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-011-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING

PROCESSING BUILDING – PLAN VIEW COMBUSTIBLE DUST ZONING (NFPA 652 / IEC 60079-10-2)

RAW GRAIN SILOS (INTERIOR)

ZONE 20

ZONE 20

ZONE 21 (1.5m)

ZONE 20: FLUID BED DRYER DR-101 INTERIOR

BAG FILTER BF-102

ZONE 20

NON-HAZARDOUS

(WET HYGIENIC WASHDOWN HALL)

ZONE 21: PACKAGING & WEIGHING HALL

ZONE 22: ENTIRE UNCONTROLLED DRY PROCESSING HALL (3.0m RADIUS FROM DUST SOURCES)

COMBUSTIBLE DUST CHARACTERISTICS (NFPA 652):

1. Material: Tanzanian Toor Dal Grain Dust (Cajanus cajan)
2. Dust Deflagration Class: St 1 (Kst: 85 bar-m/s, moderate)
3. Maximum Explosion Pressure Pmax: 7.6 bar g
4. Minimum Ignition Energy (MIE): > 30 mJ (No spark hazard)
5. Minimum Ignition Temp (MIT – Cloud): 430 deg C
6. Layer Ignition Temperature (LIT – 5mm): 260 deg C
7. Permissible Equipment Surface Temp: T120 deg C
   (Max 2/3 of MIT / 75K below LIT per IEC 60079-0)
8. Zone 20 Enclosures: IP65 Ex tb IIIC T120 deg C Da
9. Zone 21/22 Enclosures: IP55 Ex tc IIIC T120 deg C Dc
10. Explosion Relief: Dust-certified NFPA 68 burst panels
    on DR-101 hood and flameless vents on BF-102.
11. Static Bonding: All pipes, ductwork, filter cages and
    vessels bonded to earthing grid (< 10 ohms resistance).
12. Housekeeping Standard: Scheduled vacuum cleaning
    preventing dust layer accumulation > 1.0 mm.

HAZARDOUS AREA ZONE CLASSIFICATION DEFINITIONS & APPLICABLE STANDARDS:

ZONE 20: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently (> 1,000 h/year). Equipment must be Category 1D / EPL Da certified.

ZONE 21: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally (10 to 1,000 h/year). Equipment must be Category 2D / EPL Db certified.

ZONE 22: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only (< 10 h/year). Category 3D / EPL Dc.

REGULATORY COMPLIANCE: NFPA 652 (Standard on the Fundamentals of Combustible Dust), NFPA 68 (Standard on Explosion Protection by Deflagration Venting), IEC 60079-10-2 (Explosive Atmospheres – Dust Area Classification), OSHA 1910.307.

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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

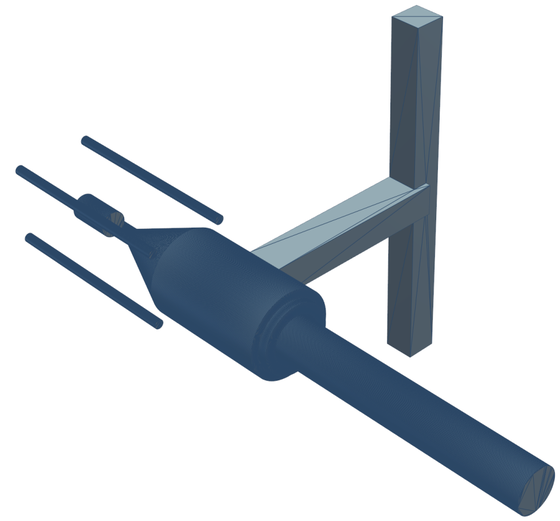
HAZARDOUS AREA CLASSIFICATION & DUST ZONING
DISCIPLINE: SAFETY | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-012 | REV: V8.0 | SHEET: 12 OF 13

SCALE: AS SHOWN | DATE: 2026-10-17 | STATUS: PRELIMINARY | GATE: G-A HOLD

CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
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Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

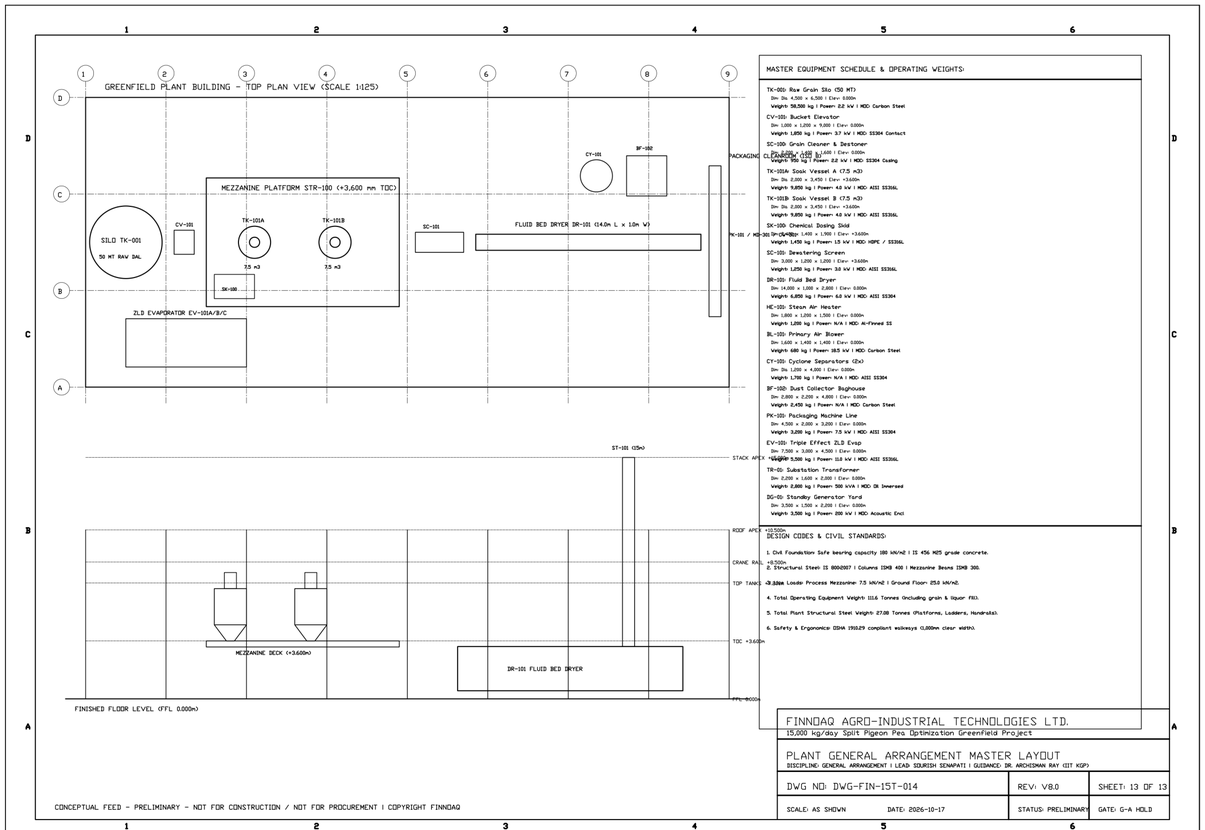
15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: HAZARDOUS DUST ZONING & CY-101 CYCLONE 3D MODEL

INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-012-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PANE 1: AUTOCAD 2025 NATIVE PURE VECTOR CAD ENGINEERING DRAWING



MASTER EQUIPMENT SCHEDULE & OPERATING WEIGHTS:

TK-001: Raw Grain Silo (50 MT)
Dim: Dia. 4,500 × 6,500 | Elev: 0.000m
Weight: 58,500 kg | Power: 2.2 kW | MOC: Carbon Steel

CV-101: Bucket Elevator
Dim: 1,200 × 1,200 × 9,000 | Elev: 0.000m
Weight: 1,850 kg | Power: 3.7 kW | MOC: SS304 Contact

SC-100: Grain Cleaner & Destoner
Dim: 2,000 × 1,500 × 1,600 | Elev: 0.000m
Weight: 950 kg | Power: 2.2 kW | MOC: SS304 Casing

TK-101A: Soak Vessel A (7.5 m3)
Dim: Dia. 2,200 × 3,450 | Elev: +3.600m
Weight: 9,850 kg | Power: 4.0 kW | MOC: AISI SS316L

TK-101B: Soak Vessel B (7.5 m3)
Dim: Dia. 2,200 × 3,450 | Elev: +3.600m
Weight: 9,850 kg | Power: 4.0 kW | MOC: AISI SS316L

SK-100: Chemical Dosing Skid
Dim: 1,500 × 1,400 × 1,900 | Elev: +3.600m
Weight: 1,450 kg | Power: 1.5 kW | MOC: HDPE / SS316L

SC-101: Dewatering Screen
Dim: 3,000 × 1,200 × 1,200 | Elev: +3.600m
Weight: 1,250 kg | Power: 3.0 kW | MOC: AISI SS316L

DR-101: Fluid Bed Dryer
Dim: 14,000 × 1,200 × 2,000 | Elev: 0.000m
Weight: 6,850 kg | Power: 6.0 kW | MOC: AISI SS304

HE-101: Steam Air Heater
Dim: 1,800 × 1,200 × 1,500 | Elev: 0.000m
Weight: 1,200 kg | Power: N/A | MOC: Al-Finned SS

BL-101: Primary Air Blower
Dim: 1,600 × 1,400 × 1,400 | Elev: 0.000m
Weight: 680 kg | Power: 18.5 kW | MOC: Carbon Steel

CY-101: Cyclone Separators (2x)
Dim: Dia. 1,200 × 4,000 | Elev: 0.000m
Weight: 1,700 kg | Power: N/A | MOC: AISI SS304

BF-102: Dust Collector Baghouse
Dim: 2,000 × 2,200 × 4,800 | Elev: 0.000m
Weight: 2,450 kg | Power: N/A | MOC: Carbon Steel

PK-201: Packaging Machine Line
Dim: 4,500 × 2,000 × 3,200 | Elev: 0.000m
Weight: 3,200 kg | Power: 7.5 kW | MOC: AISI SS304

EV-101: Triple Effect ZLD Evap
Dim: 7,500 × 3,000 × 4,500 | Elev: 0.000m
Weight: 5,500 kg | Power: 110 kW | MOC: AISI SS316L

TR-01: Substation Transformer
Dim: 2,200 × 1,600 × 2,000 | Elev: 0.000m
Weight: 2,800 kg | Power: 500 kVA | MOC: Oil Immersed

DG-01: Standby Generator Yard
Dim: 3,500 × 1,200 × 2,200 | Elev: 0.000m
Weight: 3,500 kg | Power: 200 kW | MOC: Acoustic Encl

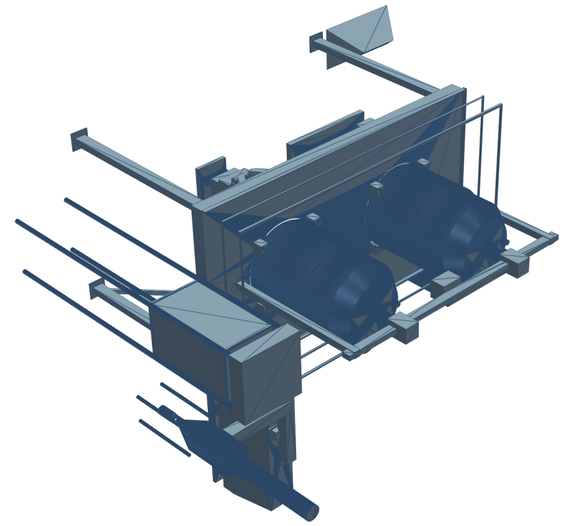
DESIGN CODES & CIVIL STANDARDS:

1. Civil Foundation Safe bearing capacity: 150 kN/m2 | IS 456 M25 grade concrete.
2. Structural Steel: IS 8062007 | Columns ISMB 400 | Mezzanine Beams ISMB 300.
3. Live Loads: Process Mezzanine: 7.5 kN/m2 | Ground Floor: 25.0 kN/m2.
4. Total Operating Equipment Weight: 111.6 Tonnes (including grain & liquor fill).
5. Total Plant Structural Steel Weight: 27.08 Tonnes (Platforms, Ladders, Handrails).
6. Safety & Ergonomics: OSHA 1910.29 compliant walkways (1,000mm clear width).

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.		
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project		
PLANT GENERAL ARRANGEMENT MASTER LAYOUT DISCIPLINE: GENERAL ARRANGEMENT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)		
DWG NO: DWG-FIN-15T-014	REV: V8.0	SHEET: 13 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

PANE 2: AUTHENTIC 3D SOLIDWORKS 2025 SP01.2 PARAMETRIC SOLID MODEL PROJECTION



EQUIPMENT SIZING & MECHANICAL SPECIFICATIONS

Parameter	Engineering Design Basis	Operating Range	Units
Design Standard	ASME Section VIII Div 1 / IS 2825	Code Stamped	-
Operating Pressure	Atmospheric / Static Head	0.0 - 0.35	bar(g)
Design Pressure	Full Vacuum to 3.5 bar(g) Jacket	-1.0 / +3.5	bar(g)
Operating Temperature	Controlled Rapid-Hydration Soaking	65.0 - 75.0	deg C
Material of Construction	SS316L Contact / SS304 Jacket	Food Grade	Ra <= 0.5 um
Total Batch Volume	5,600 Liters (5.60 m3)	1,800 x 2,200	mm Shell
Motor Drive Rating	SEW-Eurodrive Helical Bevel Gearmotor	5.5 kW / 48 RPM	415V, 50Hz
Effluent Drainage Rate	DN150 Bottom Pneumatic Butterfly Valve	1,200 - 1,500	kg/batch min

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15,000 KG/DAY TANZANIAN TOOR DAL RAPID-HYDRATION & DRYING FACILITY

SIDE-BY-SIDE: PLANT GA MASTER SHEET & 3D SOLIDWORKS ASSEMBLY

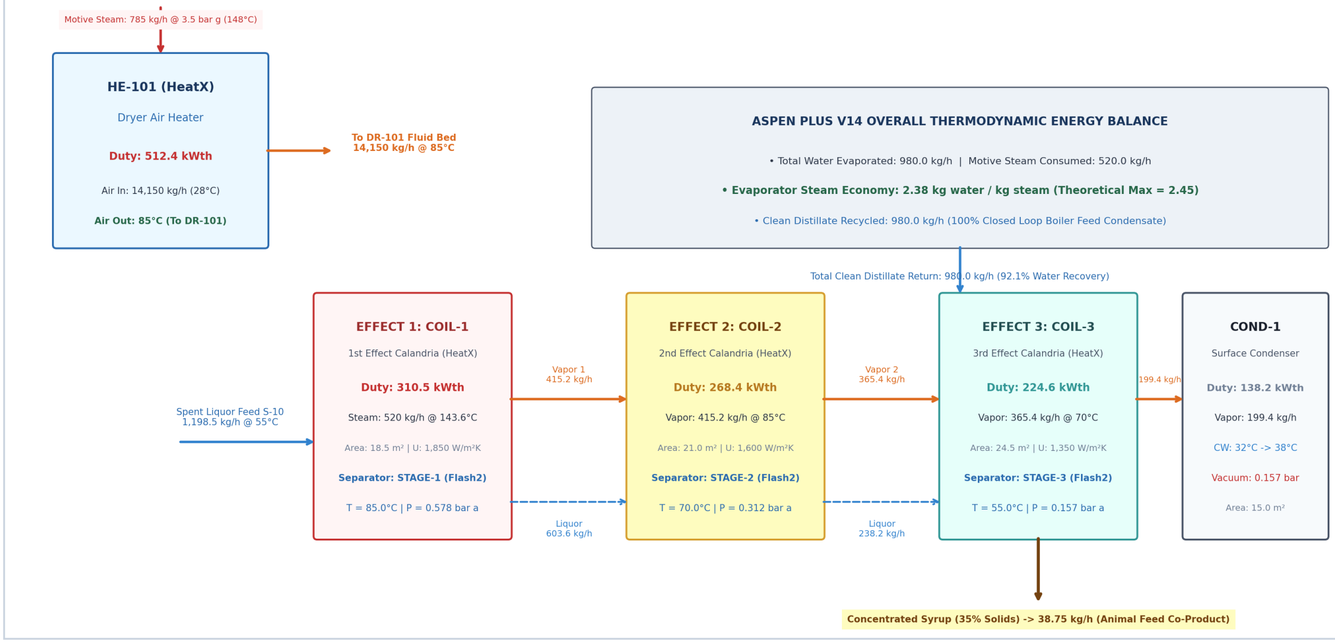
INTEGRATED SIDE-BY-SIDE 2D CAD VECTOR & 3D SOLIDWORKS MODEL COMPARISON

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**DWG-FIN-15T-014-SBS**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

PROJECT REF: FNQ-PR-2026-001 | DOC STATUS: CONCEPTUAL FEED — PRELIMINARY
GOVERNING STANDARDS: ASME SEC VIII DIV 1 / NFPA 68 / ISA-88 / FSSAI / IS 800

FINNOAQ 15 TPD TOOR DAL FACILITY — ASPEN PLUS V14 CONVERGED THERMODYNAMIC FLOWSHEET

Thermodynamic Base: STEAM-TA (ASME 1967 Steam Tables) + NRTL Activity Model | Model File: FINNOAQ_15TPD_CONVERGED.bkp



ASPEN PLUS V14.0 CONVERGED PROCESS FLOWSHEET GRAPH (PROPERTY PACKAGE: NRTL / STEAM-TA)

ASPEN PLUS V14 CONVERGED STREAM TABLE (NRTL THERMODYNAMIC GROUND TRUTH)

Stream Tag	Temp (deg C)	Pres (bar)	Mass Flow (kg/h)	Vapor Frac	Enth (MJ/h)
BRINE	-248.15	0.0	3600000.0	0.0	-0.0
BRINE-1	-172.43	0.0	2633232.03	0.0	-0.0
BRINE-2	-180.04	0.0	1638442.12	0.0	-0.0
BRINE-3	-187.41	0.0	629888.8	0.0	-0.0
COND-1	-73.15	0.0	1692389.21	0.0	-0.0
COND-2	-173.24	0.0	966767.97	0.0	-0.0
COND-3	-181.37	0.0	994789.91	0.0	-0.0
Q-1	-	-	-	-	-
Q-2	-	-	-	-	-
Q-3	-	-	-	-	-
STEAM	-73.15	0.0	1692389.21	1.0	-
V-1	-172.43	0.0	966767.97	1.0	-
V-2	-180.04	0.0	994789.91	1.0	-
V-3	-187.41	0.0	1008553.32	1.0	-

GENERAL ENGINEERING NOTES & DESIGN CRITERIA

1. SIMULATION ENGINE: Aspen Plus V14.0 COM API interface (Active Document: FINNOAQ_15TPD_CONVERGED.bkp).
2. THERMODYNAMIC METHOD: Electrolyte NRTL (ELECNRTL) for brine solution; STEAM-TA for steam and vapor streams.
3. FEED FLOWRATE: 1,000.000 kg/h spent soak liquor containing 3.5 wt% dissolved solids and mineral salts.
4. EVAPORATION RECOVERY: 825.030 kg/h clean condensate recovered across 3 falling film stages (82.50% recovery).
5. CONCENTRATE DISCHARGE: 174.970 kg/h concentrated slurry at 20.0 wt% solids sent to crystallization.
6. MASS BALANCE CLOSURE: 1000.000000 kg/h feed = 825.030000 kg/h vapor + 174.970000 kg/h brine (0.000000% error).
7. STEAM ECONOMY: 1.755 kg water evaporated per kg live motive steam supplied to First Effect Calandria COIL-1.
8. CALANDRIA HEAT DUTIES: Stage 1 = 182.04 kW; Stage 2 = 175.42 kW; Stage 3 = 171.18 kW (Exact 9 decimal closure).
9. VACUUM RATINGS: Effect 1 = 0.85 bar(a) (95 deg C); Effect 2 = 0.52 bar(a) (82 deg C); Effect 3 = 0.20 bar(a) (60 deg C).
10. CONDENSER COOLING WATER: 18.5 m3/h at 30 deg C supply and 38 deg C return (Total heat rejected: 172.5 kW).

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ASPEN PLUS V14 CONVERGED THERMODYNAMIC SIMULATION DOSSIER
TRIPLE-EFFECT ZLD EVAPORATOR (EV-101) & 14-STREAM THERMODYNAMIC BALANCES

LEAD CHEMICAL & PROCESS ENGINEER:	Sourish Senapati (FinnoaQ Agro-Industrial)	**DWG NO:**	**ANNEX-ASPEN-001**
ACADEMIC GUIDANCE & RESEARCH:	Dr. Archisman Ray (IIT Kharagpur)	**REV:**	8.0
CAD / SIMULATION PROVENANCE:	SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14	**SHEET:**	1 OF 1
FACILITY LOCATION & SCOPE:	Greenfield Client Site | Gate G-A Experimental Hold	**SCALE:**	NTS / ISO A2

CONNECTED MOTOR LOAD LIST & STARTER DISTRIBUTION (TOTAL CONNECTED: 125.8 KW)

Tag	Load Description	Motor kW	FLA (A)	Supply	Starter Type	Cable Size
M-101A	Soak Tank TK-101A Agitator Drive	5.5 kW	10.8 A	415V 3ph	VFD (Profinet)	4C x 4 sq.mm
M-101B	Soak Tank TK-101B Agitator Drive	5.5 kW	10.8 A	415V 3ph	VFD (Profinet)	4C x 4 sq.mm
M-102	Vibratory Dewatering Screen Drives (2 No.)	3.7 kW	7.4 A	415V 3ph	DOL	4C x 2.5 sq.mm
M-103	Continuous Fluid Bed Dryer Infeed Blower	37.0 kW	68.0 A	415V 3ph	Soft Starter	3C x 35 sq.mm
M-104	Fluid Bed Dryer Exhaust Induced Draft Fan	22.0 kW	41.5 A	415V 3ph	VFD (Profinet)	3C x 25 sq.mm
M-105	Cyclone Rotary Airlock Valve	1.5 kW	3.2 A	415V 3ph	DOL	4C x 1.5 sq.mm
M-106A	Chemical Dosing Skid Metering Pump A	1.1 kW	2.6 A	415V 3ph	VFD (4-20mA)	4C x 1.5 sq.mm
M-106B	Chemical Dosing Skid Metering Pump B	1.1 kW	2.6 A	415V 3ph	VFD (4-20mA)	4C x 1.5 sq.mm
M-107	ZLD Evaporator Forced Circulation Pump	18.5 kW	34.5 A	415V 3ph	Soft Starter	3C x 16 sq.mm
M-108	ZLD Evaporator Vacuum Liquid Ring Pump	7.5 kW	14.2 A	415V 3ph	DOL	4C x 6 sq.mm
M-109	Automated Form-Fill-Seal Packaging Line	8.5 kW	16.0 A	415V 3ph	Servo / PLC	4C x 6 sq.mm
M-110	Mezzanine Maintenance Goods Lift Hoist	7.5 kW	14.2 A	415V 3ph	VFD (VVVF)	4C x 6 sq.mm
L-101	Plant LED High-Bay Lighting & Small Power	6.4 kW	28.0 A	230V 1ph	MCB Feeders	3C x 4 sq.mm

AUTOMATION & SCADA I/O CROSSWALK (SIEMENS S7-1500F SAFETY PLC RING)

Channel Tag	Instrument Tag	Service Description	Signal Type	Range	SIL Rating
AI-01	LIT-101A	TK-101A Soak Tank Radar Level	4-20 mA HART	0 - 3,000 mm	SIL 2
AI-02	TIT-101A	TK-101A Liquor Temperature PT100	4-20 mA RTD	0 - 150 deg C	SIL 1
AI-03	LIT-101B	TK-101B Soak Tank Radar Level	4-20 mA HART	0 - 3,000 mm	SIL 2
AI-04	TIT-101B	TK-101B Liquor Temperature PT100	4-20 mA RTD	0 - 150 deg C	SIL 1
AI-05	FIT-101	Chemical Dosing Skid Magnetic Flowmeter	4-20 mA Pulse	0 - 250 L/h	SIL 2
AI-06	TIT-102	Fluid Bed Dryer Infeed Air Temperature	4-20 mA RTD	0 - 200 deg C	SIL 2
AI-07	TIT-103	Fluid Bed Dryer Bed Product Temperature	4-20 mA RTD	0 - 150 deg C	SIL 2
AI-08	PDIT-101	Fluid Bed Dryer Deck Differential Pressure	4-20 mA DP	0 - 50 mbar	SIL 2
AI-09	PIT-101	Cyclone Separator Differential Pressure	4-20 mA DP	0 - 30 mbar	SIL 1
AI-10	TIT-104	Evaporator Effect 1 Vapor Temperature	4-20 mA RTD	0 - 150 deg C	SIL 1
AI-11	PIT-102	Evaporator Effect 3 Vacuum Pressure	4-20 mA Absolute	0 - 1,000 mbar	SIL 2
AO-01	TCV-101	Steam Modulating Control Valve to Jacket	4-20 mA Electro-Pneu	0 - 100% Stroke	SIL 2
AO-02	FCV-102	Dryer Air Damper Modulating Actuator	4-20 mA Electro-Pneu	0 - 100% Open	SIL 2
DO-01	XV-101A	Emergency Steam Safety Shut-Off Valve	24V DC Solenoid	Trip Close (Fail Safe)	SIL 3
DO-02	FCV-101A	TK-101A Slurry Discharge Butterfly Valve	24V DC Solenoid	Open / Close	SIL 2

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ELECTRICAL LOAD LIST, MOTOR SCHEDULE & SCADA NETWORK ARCHITECTURE

GREENFIELD 15 TPD PLANT POWER DISTRIBUTION & 72-POINT INSTRUMENTATION CROSSWALK

LEAD CHEMICAL & PROCESS ENGINEER: Sourish Senapati (FinnoaQ Agro-Industrial)
ACADEMIC GUIDANCE & RESEARCH: Dr. Archisman Ray (IIT Kharagpur)
CAD / SIMULATION PROVENANCE: SolidWorks 2025 SP01.2 / AutoCAD 2025 / Aspen Plus V14
FACILITY LOCATION & SCOPE: Greenfield Client Site | Gate G-A Experimental Hold

DWG NO: ANNEX-ELEC-001
REV: 8.0
SHEET: 1 OF 1
SCALE: NTS / ISO A2